

SOCIETY FOR ECOSYSTEM RESTORATION
IN NORTHERN BRITISH COLUMBIA

Restoring Fish Passage in the Peace Region - 2025

PEA-F24-F-3944-DCA - NEED TO UPDATE

**Prepared for
Fish and Wildlife Compensation Program
and
Fish Passage Technical Working Group**

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New Graph Environment Ltd.
on behalf of
Society for Ecosystem Restoration in Northern BC**

**Prepared with financial support of the Fish and Wildlife Compensation Program on
behalf of its program partners BC Hydro, the Province of BC, Fisheries and Oceans
Canada, First Nations and Public Stakeholders.**

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Acknowledgement

At New Graph Environment, we understand our well-being as inseparable from the health of the land and waters we work within. When we care for ecosystems, we care for ourselves and for the communities connected to them. This relationship is not metaphorical — it is the foundation of our practice.

Modern civilization has a long journey ahead to acknowledge and address the historic and ongoing impacts of colonialism that have resulted in harm to the cultures and livelihoods living interconnected with our ecosystems for many thousands of years.

This work takes place within the territories of [Nations], who have stewarded these lands and waters since time immemorial. [Project-specific connection to territory, governance, species, or watershed.]

[Funding and partner acknowledgements.]

AI Disclosure

Report preparation was assisted by Claude (Anthropic), a large language model, for literature synthesis and technical writing. All ecological assessments, professional judgments, and conclusions are the sole responsibility of Al Irvine, R.P.Bio., who independently verified all content.

Executive Summary

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_peace_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest version stamped PDF from [fish_passage_peace_2024_reporting.pdf](#).

The Society for Ecosystem Restoration in Northern BC (SERNbc) is working together with the McLeod Lake Indian Band, the Peace Region Fish and Wildlife Compensation Program (FWCP), the Provincial Fish Passage Technical Working Group (FPTWG), road/rail tenure holders and other stakeholders/partners to prioritize, plan and fund the restoration of fish passage at road crossing structure barriers within the Parsnip River, Carp River and Crooked River watershed groups.

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

Although the main purpose of this report is to document field work data and results, it also builds on reporting from past field activities and all reports can be considered living documents that can be updated and improved over time. In addition to the numerous assessments at sites undocumented in past years of the project, field activities in 2024 were also conducted at sites where habitat confirmations were previously documented within the reports linked below. The reports for these sites were edited and updated with data.

- [Parship River Watershed – Fish Habitat Confirmations \(PEA-F20-F-2967\)](#) (Irvine 2020)
- [PEA-F22-F-3577-DCA Restoring Fish Passage in the Peace Region](#) (Irvine 2022)
- [Restoring Fish Passage in the Peace Region - 2022 - PEA-F23-F-3761-DCA](#) (Irvine and Winterscheidt 2023)
- [Restoring Fish Passage in the Peace Region - 2023](#) (Irvine and Winterscheidt 2024)
- [Restoring Fish Passage in the Peace Region - 2024](#) (Irvine and Schick 2025)

Fish passage assessment procedures conducted through SERNbc in the Upper Fraser River Watershed since 2023 are amalgamated online within the Results and Discussion section of the report found [here](#) which includes links to project reporting for each site.

Executive Summary

In 2025, the project area was expanded to include the Willow river, Lower Salmon River, and Tabor River watershed groups. Fish passage assessments were completed at 12 new sites, focusing on structures with potential barriers to upstream fish movement using standard provincial criteria.

Habitat confirmation assessments were conducted at multiple sites within the Willow river and Tabor River watersheds, covering over XX km of stream length. Detailed habitat metrics were documented alongside eDNA samples to evaluate habitat quality and presence of fish species.

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 35 samples were collected across 10 streams.

Monitoring was conducted at two sites, including post-remediation evaluations at PSCIS crossings 196085 and 196200. Monitoring included eDNA sampling and habitat observations. A custom effectiveness monitoring form tailored to fish passage projects was used and metrics assessed included flow velocity, substrate condition, channel constriction, riparian condition, and cover availability.

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices—picking battles carefully, building trust, and staying committed to a longer-term transformation.

While preliminary top remediation priorities are provided by watershed group, these rankings are inherently subjective and can depend on the capacity and willingness of infrastructure owners and tenure holders to support implementation—both financially and over the often multi-year project timelines. In practice, we must often act opportunistically, pursuing simpler, lower-cost options to maintain momentum and achieve near-term progress.

NEEDS TO BE UPDATED

To enhance fish passage restoration in the FWCP Peace Region:

- Maintain strong partnerships to support funding, site selection, remediation, and monitoring through adaptive management informed by traditional knowledge and real-time data.
- Prioritize detailed assessments in areas with blockages and high habitat potential, especially near McLeod Lake.
- Use climate modeling to prioritize crossings that enable access to cold, drought-resistant habitats.
- Secure financial commitments for Fern Creek remediation despite uncertainties in harvest planning.
- Continue effectiveness monitoring at key sites using fish sampling, eDNA, PIT tagging, temperature data, and aerial imagery.
- Continue to develop a cost-effective monitoring framework to assess productivity gains from improved passage.
- Collaborate with WLRS, UNC, local fisheries experts, FWCP, and the CEMPRA Project working group.
- Utilize environmental DNA (eDNA) to better understand bull trout and Arctic grayling habitat use at both potential and remediated sites. This will refine prioritization and assess fish passage effectiveness.

1 Introduction

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This project builds on Society for Ecosystem Restoration Northern BC (SERNbc) work in:

- [Parship River Watershed – Fish Habitat Confirmations \(PEA-F20-F-2967\)](#) (Irvine 2020)
- [PEA-F22-F-3577-DCA Restoring Fish Passage in the Peace Region](#) (Irvine 2022)
- [Restoring Fish Passage in the Peace Region - 2022 - PEA-F23-F-3761-DCA](#) (Irvine and Winterscheidt 2023)
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The health and viability of freshwater fish populations can depend on access to tributary and off channel areas which provide refuge during high flows, opportunities for foraging, overwintering habitat, spawning habitat and summer rearing habitat (Bramblett et al. 2002; Swales and Levings 1989; Diebel et al. 2015). Culverts can present barriers to fish migration due to low water depth, increased water velocity, turbulence, a vertical drop at the culvert outlet and/or maintenance issues (Slaney, Zaldokas, and Watershed Restoration Program (B.C.) 1997; Cote et al. 2005). As road crossing structures are commonly upgraded or removed there are numerous opportunities to restore connectivity by ensuring that fish passage considerations are incorporated into repair, replacement, relocation and deactivation designs.

1 Introduction

Although remediation and replacement of stream crossing structures can have benefits to local fish populations, the costs of remedial works can be significant and the impacts of the work often complex to evaluate and quantify. Additionally, allocation of ecosystem restoration funding towards infrastructure upgrades on transportation right of ways are not always considered ethical under all circumstances from all perspectives. When funds are finite and invested groups are engaged in fund raising, cost benefits and the ethics of crossing replacements should be explored collaboratively alongside the cost benefits and ethics of alternative investment activities including transportation corridor relocation/deactivation, land procurement/covenant, cattle exclusion, riparian/floodplain restoration, habitat complexing, water conservation, commercial/recreational fishing management, salt water interventions and research.

Please note that at the time of reporting this document can be considered a living document. See the [Open Source Reporting Framework \(page 29\)](#) section for details on version tracking and project documentation.

2 Background

2.1 Project Location

The study area includes the FWCP Peace Region with a focus to date on traditional territories of the Tse'khene First Nations. In 2025, field assessments were completed with the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups (Figure [2.1](#)).

In 2019/2020, following a literature review, analysis of fish habitat modelling data, the Provincial Stream Crossing Inventory System (PSCIS) and a community scoping exercise within the McLeod Lake Indian Band habitat confirmation assessments were conducted at 17 sites throughout the Parsnip River watershed with 10 crossings rated as high priorities for rehabilitation and three crossings rated as moderate priorities for restoration. An engineering design for site 125179 on a tributary to the Missinka River was also completed through the 2019/2020 project. In 2021/2022, project activities reconvened through FWCP directed project PEA-F22-F-3577-DCA. Partners were engaged, funding was raised, planning was conducted and reporting was completed to initiate restoration activities of high priority crossings. Materials were purchased and permitting was put in place to prepare for replacement of the twin culverts on the Missinka River tributary with a clear-span bridge. In 2025/2026, this collaborative project leveraged ongoing connectivity restoration initiatives in the province and engaged multiple partners to catalyze fish passage restoration activities at high-priority sites identified in from 2019-2024.

2 Background

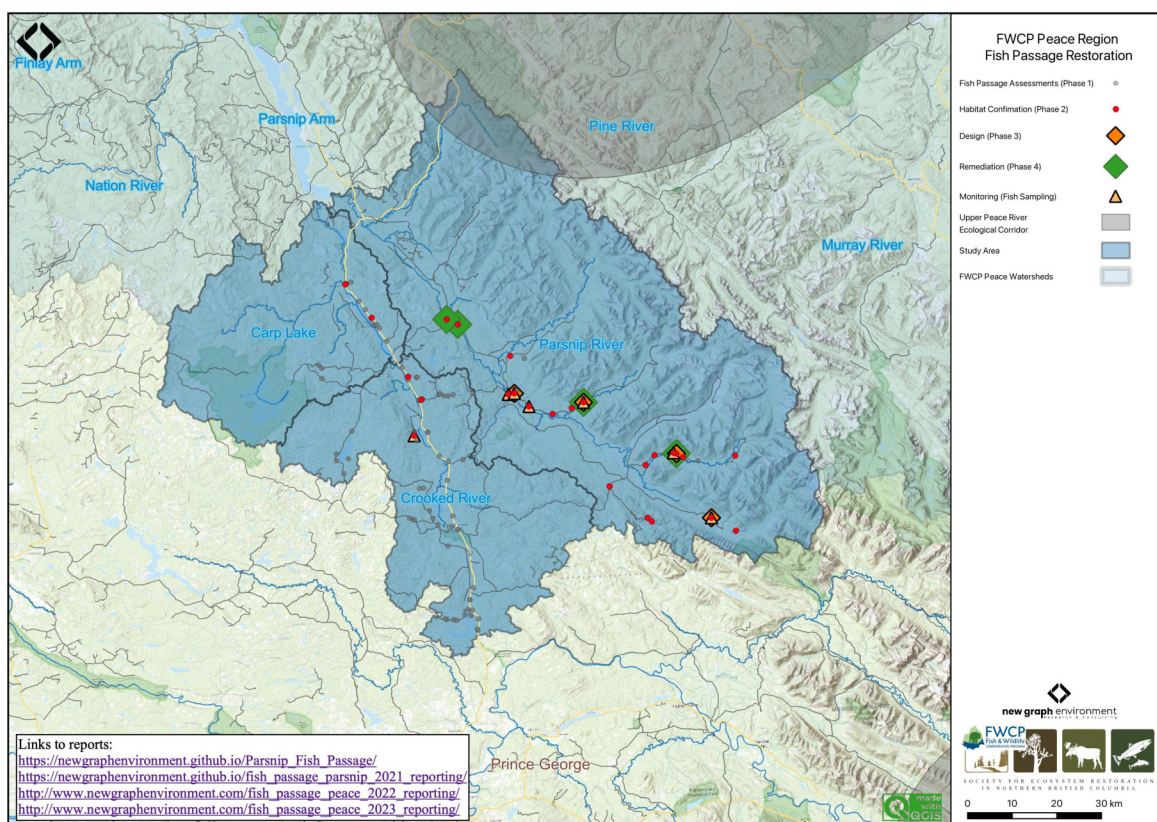


Figure 2.1: Overview map of Study Area

2.2 Tse'khene

The Parsnip River watershed is located within the south-eastern portion of the 108,000 km² traditional territory of the Tse'khene from the [McLeod Lake Indian Band](#). The Tse'khene "*People of the Rocks*" are a south westerly extension of the Athabaskan speaking people of northern Canada. They were nomadic hunters whose language belongs to the Beaver-Sarcee-Tse'khene branch of Athapaskan ("History Who We Are" 2023). Extensive work is underway to preserve the language with resources such as First Voices available [online](#) and in [app form](#) for iphone and ipad devices.

The continental divide separates watersheds flowing north into the Arctic Ocean via the Mackenzie River and south and west into the Pacific Ocean via the Fraser River (Figure 2.1). The Parsnip River is a 6th order stream with a watershed that drains an area of 5597km². The mainstem of the river flows within the Rocky Mountain Trench in a north direction into Williston Reservoir starting from the continental divide adjacent to Arctic Lakes. Major tributaries include the Misinchinka, Colbourne, Reynolds, Anzac, Table, Hominka and Missinka sub-basins which drain the western slopes of the Hart Ranges of the Rocky Mountains. The Parsnip River has a mean annual discharge of 150.5 m³/s with flow patterns typical of high elevation watersheds on the west side of

2.2 Tse'khene

the northern Rocky Mountains which receive large amounts of precipitation as snow leading to peak levels of discharge during snowmelt, typically from May to July (Figure 2.2).

Construction of the 183m high and 2134m long W.A.C. Bennett Dam was completed in 1967 at Hudson's Hope, BC, creating the Williston Reservoir (Hirst 1991). Filling of the 375km² reservoir was complete in 1972 and flooded a substantial portion of the Parsnip River and major tributary valleys forming what is now known as the Peace and Parsnip reaches. The replacement of riverine habitat with an inundated reservoir environment resulted in profound changes to the ecology, resource use and human settlement patterns in these systems (Hagen et al. 2015a; Pearce et al. 2019; Stamford, Hagen, and Williamson 2017). Prior to the filling of the reservoir, the Pack River, into which McLeod Lake flows, was a major tributary to the Parsnip River. The Pack River currently enters the Williston Reservoir directly as the historic location of the confluence of the two rivers lies within the reservoir's footprint.

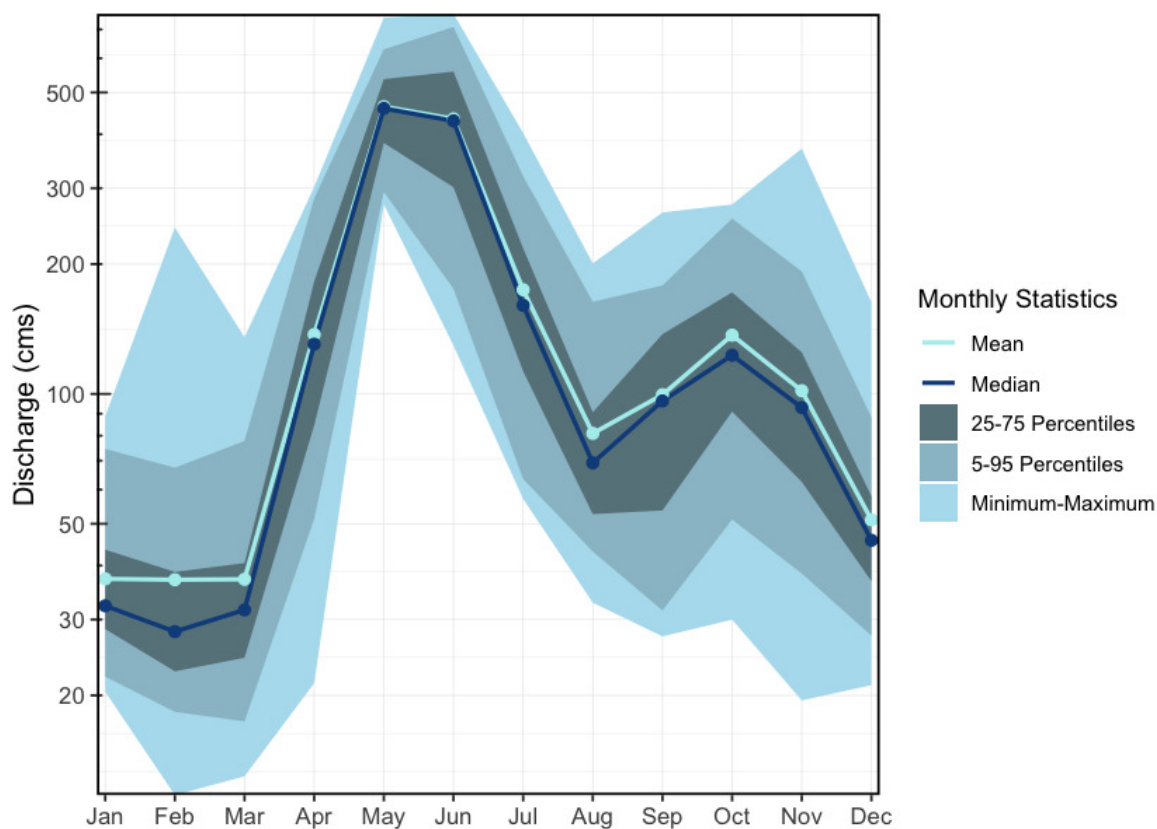


Figure 2.2: Hydrograph for Parsnip River above Misinchinka River (Station #07EE007).

2 Background

2.3 Fisheries

Fish species recorded in the BC Fisheries Information Summary System (FISS) in the Parsnip River watershed are detailed in Table 2.1 (MoE 2019). In addition to flooding related to the formation of the Williston Reservoir, transmission lines, gas pipelines, rail, road networks, forestry, elevated water temperatures, interrupted connectivity, invasion from non-native species and insect infestations affecting forested areas pose threats to fisheries values in the Parsnip River watershed (Hagen et al. 2015b; Stamford, Hagen, and Williamson 2017; Hagen and Weber 2019a; Committee on the Status of Endangered Wildlife in Canada 2012). A brief summary of trends and knowledge status related to Arctic grayling, bull trout, kokanee, mountain whitefish and rainbow trout in Williston Watershed streams is provided in Fish and Wildlife Compensation Program (2020) with a more detailed review of the state of knowledge for Parsnip River watershed populations of Arctic grayling and bull trout provided below.

Table 2.1: Fish species recorded in FISS in the Parsnip River, Carp Lake, and Crooked River watershed groups.

Scientific Name	Species Name	BC List	COSEWIC	Carp Lake	Crooked	Nation	Parship Arm	Parship
<i>Catostomus catostomus</i>	Longnose Sucker	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Catostomus columbianus</i>	Bridgelip Sucker	Yellow	–	–	Yes	–	–	–
<i>Catostomus commersonii</i>	White Sucker	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Catostomus macrocheilus</i>	Largescale Sucker	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Coregonus clupeaformis</i>	Lake Whitefish	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Cottus aleuticus</i>	Coastrange Sculpin (formerly Aleutian Sculpin)	Yellow	–	–	–	–	–	Yes
<i>Cottus asper</i>	Prickly Sculpin	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Cottus cognatus</i>	Slimy Sculpin	Yellow	–	Yes	–	Yes	Yes	Yes
<i>Cottus ricei</i>	Spoonhead Sculpin	Yellow	NAR (May 1989)	–	–	–	Yes	–
<i>Couesius plumbeus</i>	Lake Chub	Yellow	DD	Yes	Yes	Yes	Yes	Yes
<i>Esox lucius</i>	Northern Pike	Yellow	–	–	Yes	–	–	–
<i>Hybognathus hankinsoni</i>	Brassy Minnow	No Status	–	Yes	Yes	Yes	Yes	–
<i>Lota lota</i>	Burbot	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Mylocheilus caurinus</i>	Peamouth Chub	Yellow	–	Yes	Yes	Yes	Yes	Yes

2.3 Fisheries

Scientific Name	Species Name	BC List	COSEWIC	Carp Lake	Crooked	Nation	Parsnip Arm	Parsnip
<i>Oncorhynchus mykiss</i>	Rainbow Trout	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Oncorhynchus nerka</i>	Kokanee	Not Reviewed	–	–	–	Yes	Yes	Yes
<i>Osmerus dentex</i>	Rainbow Smelt	Unknown	–	–	–	Yes	–	Yes
<i>Prosopium coulterii</i>	Pygmy Whitefish	Yellow	NAR (Nov 2016)	–	–	Yes	Yes	Yes
<i>Prosopium cylindraceum</i>	Round Whitefish	Yellow	–	–	–	–	–	Yes
<i>Prosopium williamsoni</i>	Mountain Whitefish	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Ptychocheilus oregonensis</i>	Northern Pikeminnow	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Rhinichthys cataractae</i>	Longnose Dace	Yellow	–	–	Yes	Yes	Yes	Yes
<i>Richardsonius balteatus</i>	Redside Shiner	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Salvelinus confluentus</i>	Bull Trout	Blue	SC (Nov 2012)	Yes	Yes	Yes	Yes	Yes
<i>Salvelinus fontinalis</i>	Brook Trout	Exotic	–	–	Yes	–	Yes	Yes
<i>Salvelinus malma</i>	Dolly Varden	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Salvelinus namaycush</i>	Lake Trout	Yellow	–	Yes	Yes	Yes	Yes	Yes
<i>Thymallus arcticus</i>	Arctic Grayling	Yellow	–	–	–	Yes	Yes	Yes
–	Chub (General)	–	–	Yes	Yes	–	Yes	Yes
–	Dace (General)	–	–	–	–	–	Yes	Yes
–	Lamprey (General)	–	–	–	–	Yes	–	–
–	Minnow (General)	–	–	Yes	Yes	–	Yes	Yes
–	Mottled Sculpin	–	–	–	–	Yes	–	Yes
–	Salmon (General)	–	–	–	–	Yes	–	–
–	Sculpin (General)	–	–	Yes	Yes	Yes	Yes	Yes
–	Smelt (General)	–	–	–	–	–	Yes	–
–	Squanga	–	–	–	Yes	–	–	–
–	Sucker (General)	–	–	Yes	Yes	Yes	Yes	Yes
–	Whitefish (General)	–	–	Yes	Yes	Yes	Yes	Yes

* COSEWIC abbreviations :

SC - Special concern

DD - Data deficient

NAR - Not at risk

BC List definitions :

Yellow - Species that is apparently secure

Blue - Species that is of special concern

Exotic - Species that have been moved beyond their natural range as a result of human activity

2 Background

Scientific Name	Species Name	BC List	COSEWIC	Carp Lake	Crooked	Nation	Parsnip Arm	Parsnip
Dolly Varden are documented in all three study watersheds—Carp, Crooked, and Parsnip—but these records are likely misidentifications of bull trout (Committee on the Status of Endangered Wildlife in Canada 2012).								

Bull Trout - sa'ba

Tse'khene Elders from the McLeod Lake Indian Band report that sa'ba (bull trout) size and abundance has decreased in all rivers and tributaries from the reservoir with more injured and diseased fish captured in recent history than was common in the past (Pearce et al. 2019) .

Bull trout populations of the Williston Reservoir watershed are included within the Western Arctic population 'Designatable Unit 10', which, in 2012, received a ranking of 'Special Concern' by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC 2012). They were added to Schedule 1 under the Species at Risk Act in 2019 (Species Registry Canada 2020) and are also considered of special concern (blue-listed) provincially (BC Species & Ecosystem Explorer 2020). Some or all of the long-term foot survey index sections of four Williston Reservoir spawning tributaries (Davis Creek, Misinchinka River, Point Creek, and Scott Creek), have been surveyed within 16 of the 19 years between 2001 and 2019 (16 of 19 in Davis River, 10 years over a 13-year period in the Misinchinka River, 11 years over a 14-year period for Point Creek, and 9 years over an 11-year period for Scott Creek (Hagen, Spendlow, and Pillipow 2020).

A study of bull trout critical habitats in the Parsnip River was conducted in 2014 with the Misinchinka and Anzac systems identified as the most important systems for adfluvial (large bodied) bull trout spawners. The Table River was also highlighted as an important spawning destination. Other watersheds identified as containing runs of large bodied bull trout spawners included the Colbourne, Reynolds, Hominka and Missinka River with potentially less than 50 spawners utilizing each sub-basin (Hagen et al. 2015a). Hagen and Weber (2019b) have synthesized a large body of information regarding limiting factors, enhancement potential, critical habitats and conservation status for bull trout of the Williston Reservoir and the reader is encouraged to review this work for context. They have recommended experimental enhancements within a monitoring framework for Williston Reservoir bull trout (some spawning and rearing in Parsnip River mainstem and tributaries) which include stream fertilization, side channel development, riparian restoration and fish access improvement.

In 2018, sub-basins of the Anzac River watershed, Homininka River, Missinka River and Table River watersheds were designated as fisheries sensitive watersheds under the authority of the *Forest and Range Practices Act* due to significant downstream fisheries values and significant watershed sensitivity (Beaudry 2013a, 2014a, 2014b, 2013b). Special management is required in these watersheds to protect habitat for fish species including bull trout and Arctic grayling including measures (among others) to limit equivalent clearcut area, reduce impacts to natural stream

2.3 Fisheries

channel morphology, retain old growth attributes and maintain fish habitat/movement (Forest and Range Practices Act 2018).

Arctic Grayling - dusk'ihje

Tse'khene Elders from the McLeod Lake Indian Band report that dusk'ihje (Arctic grayling) numbers have declined dramatically since the flooding of the reservoir and that few dusk'ihje have been caught in the territory in the past 30 years (Pearce et al. 2019).

Since impoundment of the Williston Reservoir, it appears that physical habitat and ecological changes have been the most significant factors limiting Arctic grayling productivity. Although these changes are not well understood they have likely resulted in the inundation of key low gradient juvenile rearing and overwintering habitats, isolation of previously connected populations and increases in abundance of predators such as bull trout (Shrimpton, Roberts, and Clarke 2012; Hagen, Pillipow, and Gantner 2018). Rapid increases in industrial activity and angler access in the Parsnip River watershed pose significant risks to Arctic Grayling productivity with these threats primarily linked to forestry and pipeline initiatives (Hagen and Stamford 2021).

A detailed review of Arctic grayling life history is available in Stamford, Hagen, and Williamson (2017). Migration of mature adult Arctic grayling occurs in the spring, typically coinciding with water temperatures of 4°C. In the Parsnip watershed, spawning appears to occur between late May and late June, primarily within the lower reaches of the Anzac and Table rivers and the Parsnip River mainstem. Spawning habitat is generally associated with side-channel and multi-channel areas containing small gravels. The current distribution of Williston Arctic grayling is concentrated in fourth-order and larger streams (Williamson and Zimmerman 2005; Stamford, Hagen, and Williamson 2017). According to Stewart et al. (2007), Arctic grayling spawn in both large and small tributaries to rivers and lakes, intermittent streams, mainstem rivers, and lakes—most commonly at tributary mouths. Although earlier studies suggest that 0+ grayling overwinter in the lower reaches of larger tributaries (e.g., Table and Anzac rivers) and the Parsnip River, and that few age-1+ grayling have been sampled in tributaries, habitat use in small tributaries and the extent to which they are connected with mainstem habitats across all core areas remains poorly understood. Between 1995 and 2019, population monitoring of Arctic grayling was conducted in the Table River in 9 out of 25 years, and in the Anzac River for 8 years, using snorkel surveys. Results from 2018 and 2019 support efforts to assess the conservation status of the species in the Parsnip Core Area (Hagen, Pillipow, and Gantner 2018). Preliminary telemetry data from 2019 indicate that both Arctic grayling and bull trout rely on the Parsnip River mainstem for overwinter residency. Arctic grayling typically begin moving into tributaries in April and are widespread across the watershed by June, with larger adults observed in headwater habitats by August.

A five-year study on Arctic grayling abundance and trends in the Parsnip River watershed is presented in Hagen and Stamford (2023), which reports that the most productive summer rearing

2 Background

habitats for Arctic grayling are within the Anzac and Table rivers. Although estimated abundance is lower than in the Anzac and Table, productive summer rearing habitats for adult Arctic grayling in the upper Parsnip River watershed are distributed between 36–25km of the Missinka River and 48–32km of the Hominka River. In the Anzac River, Hagen and Stamford (2021) report that a 30km stretch from 47km to 16km supports high Arctic grayling abundance and is assumed to provide productive rearing habitat for adults. A chute obstruction located at 47km likely limits further upstream movement. Although the spatial distribution of high Arctic grayling abundance in the Table River has not been delineated through reconnaissance surveys, it has been observed to span at least a 20km zone from the waterfall migration barrier at 37km to 18km.

Spatial ecology studies in the Parsnip River between 2018 and 2021 has been reported on by Martins et al. (2022) with results related to:

- temperature modeling and spatio-temporal patterns in thermal habitat,
- telemetry data modeling and arctic grayling spatial ecology, and
- trophic relationships between Arctic grayling and bull trout

Fish presence by Habitat Type

A review of available fisheries data for the Parsnip River watershed stratified by different habitat characteristics (stream gradient, channel width, and watershed size) can provide insight into which habitats may provide the highest intrinsic value for fish species based on the number of fish captured in those habitats in past assessment work (Tables [2.2](#) - [2.4](#) and Figures [2.3](#) - [2.5](#)). It should be noted however that it should not be assumed that all habitat types have been sampled in a non-biased fashion or that particular sites selected do not have a disproportionate influence on the overall dataset composition (ie. fish salvage sites are often located adjacent to construction sites which are more commonly located near lower gradient stream reaches).

Table 2.2: Summary of historic fish observations vs. stream gradient category for the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups.

species_code	Gradient	Count	total_spp	Percent
BT	0 - 3 %	411	614	67
BT	03 - 5 %	86	614	14
BT	05 - 8 %	58	614	9
BT	08 - 15 %	50	614	8
BT	15 - 22 %	9	614	1
GR	0 - 3 %	249	251	99

2.3 Fisheries

species_code	Gradient	Count	total_spp	Percent
GR	03 - 5 %	2	251	1
KO	0 - 3 %	145	150	97
KO	03 - 5 %	2	150	1
KO	05 - 8 %	2	150	1
KO	22+ %	1	150	1
RB	0 - 3 %	2035	2619	78
RB	03 - 5 %	275	2619	11
RB	05 - 8 %	186	2619	7
RB	08 - 15 %	103	2619	4
RB	15 - 22 %	14	2619	1
RB	22+ %	6	2619	0

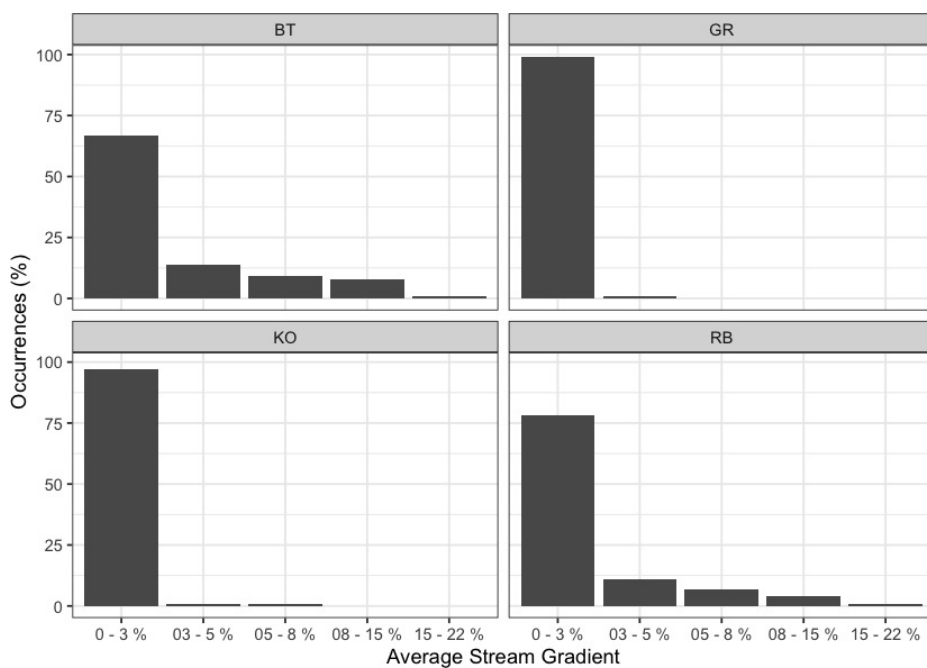


Figure 2.3: Summary of historic fish observations vs. stream gradient category for the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups.

2 Background

Table 2.3: Summary of historic fish observations vs. channel width category for the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups.

species_code	Width	Count	total_spp	Percent
BT	0 - 2m	30	614	5
BT	02 - 04m	96	614	16
BT	04 - 06m	81	614	13
BT	06 - 10m	82	614	13
BT	10 - 15m	59	614	10
BT	15m+	248	614	40
BT	–	18	614	3
GR	02 - 04m	6	251	2
GR	06 - 10m	6	251	2
GR	10 - 15m	6	251	2
GR	15m+	231	251	92
GR	–	2	251	1
KO	0 - 2m	4	150	3
KO	02 - 04m	22	150	15
KO	04 - 06m	3	150	2
KO	06 - 10m	11	150	7
KO	10 - 15m	5	150	3
KO	15m+	31	150	21
KO	–	74	150	49
RB	0 - 2m	656	2619	25
RB	02 - 04m	520	2619	20
RB	04 - 06m	257	2619	10
RB	06 - 10m	222	2619	8
RB	10 - 15m	122	2619	5
RB	15m+	454	2619	17
RB	–	388	2619	15

2.3 Fisheries

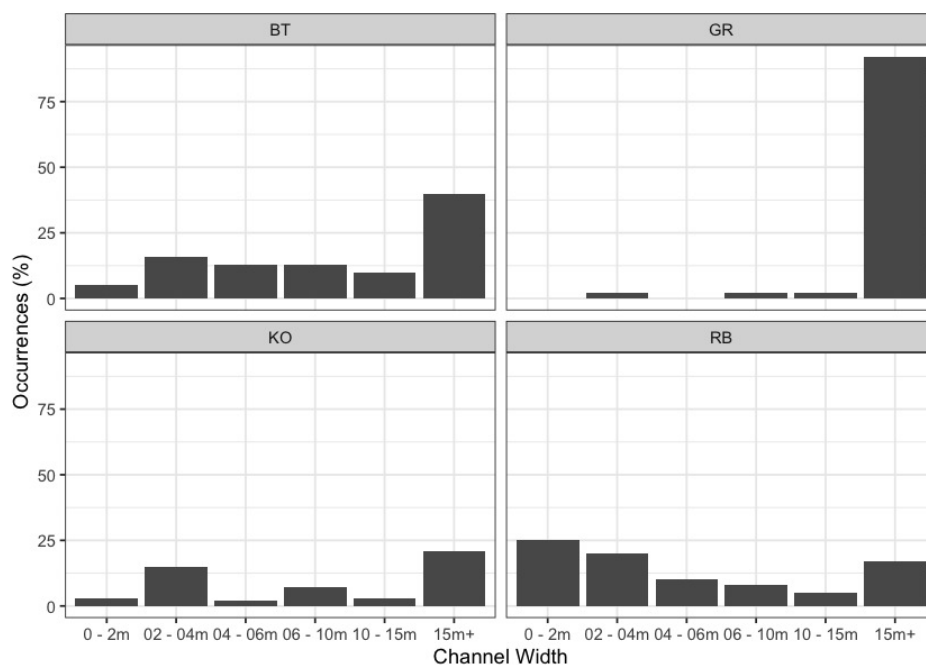


Figure 2.4: Summary of historic fish observations vs. channel width category for the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups.

Table 2.4: Summary of historic fish observations vs. watershed size category for the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups.

species_code	Watershed	count_wshd	total_spp	Percent
BT	0 - 25km2	252	614	41
BT	25 - 50km2	93	614	15
BT	50 - 75km2	40	614	7
BT	75 - 100km2	11	614	2
BT	100km2+	218	614	36
GR	0 - 25km2	8	251	3
GR	50 - 75km2	2	251	1
GR	75 - 100km2	4	251	2
GR	100km2+	237	251	94
KO	0 - 25km2	105	150	70
KO	25 - 50km2	6	150	4
KO	50 - 75km2	5	150	3

2 Background

species_code	Watershed	count_wshd	total_spp	Percent
KO	75 - 100km2	3	150	2
KO	100km2+	31	150	21
RB	0 - 25km2	1802	2619	69
RB	25 - 50km2	171	2619	7
RB	50 - 75km2	97	2619	4
RB	75 - 100km2	48	2619	2
RB	100km2+	501	2619	19

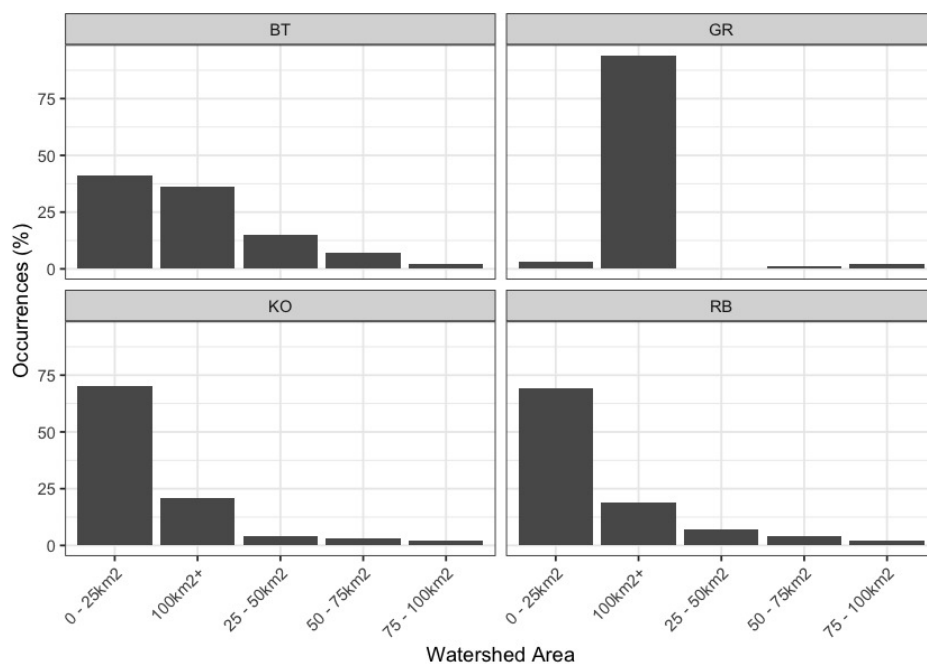


Figure 2.5: Summary of historic fish observations vs. watershed size category for the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups.

3 Methods

3.1 Collaborative GIS Environment

Geographical Information Systems are essential for developing and communicating restoration plans as well as the reasons they are required and how they are developed. Without the ability to visualize the landscape and the data that is used to make decisions, it is difficult to conduct and communicate the need for restoration, the details of past and future plans as well as and the potential results of physical works.

To facilitate the planning and implementation of restoration activities, a collaborative GIS environment has been established using [QGIS](#) and is served on the cloud using source code stored [here](#). This environment is intended to be a space where project team members can access, view, and contribute to the amalgamation of background spatial data and the development of restoration as well as monitoring for the project. The collaborative GIS environment allows users to view, edit, and analyze shared, up to date spatial data on personal computers in an office setting as well as on phones and tablets in the field. At the time of reporting, the environment was being used to develop and share maps, conduct spatial analyses, communicate restoration plans to stakeholders as well as to provide a central place to store methodologies and tools for conducting field assessments on standardized pre-developed digital forms. The platform can also be used to track the progress of restoration activities and monitor changes in the landscape over time, helping encourage the record keeping of past and future restoration activities in a coordinated manner.

The shared QGIS project was created using scripts currently kept in [dff-2022](#) with the precise calls to project creation scripts tracked in the `project_creation_and_permissions.txt` document kept in the main QGIS project directory. Information about the scripts used for GIS project creation and updates can be viewed [here](#) with outcomes of their use summarized below:

- Download and clip user specified layers from the [BC Data Catalogue](#) as well as data layers stored in custom Amazon Web Services buckets for an area of interest defined by a list of watershed groups and load to a geopackage called `background_layers.gpkg` stored in the main directory of the project.
- A project directory is created to hold the spatial data and QGIS project information (ie. layer symbology and naming conventions, metadata, etc.).
- Metadata for individual project spatial layers is kept in the `rfp_tracking` table within the `background_layers.gpkg` along with tables related to user supplied stream width/gradient inputs to `bcfishpass` to model potentially high value habitat that is accessible to fish species of interest.

3.1.1 Mapping

The workflows to produce the georeferenced pdf maps include using a QGIS layer file defining and symbolizing all layers required and are continuously evolving. At the time of reporting - mapping scripts and associated layer file were kept under version control within `bcfishpass` [here](#). Loading

the QGIS layer file within a QGIS project, allows load and representation of all map component layers provided the user points to a postgresql database populated via `bcfishpass` outputs.

3.2 Planning

3.2.1 Habitat Modelling

Habitat modelling used to help guide planning for field assessments is generated by `bcfishpass` (Norris [2020] 2024) which has been designed to prioritize potential fish passage barriers for assessment or remediation by generating a simple model of aquatic habitat connectivity. We utilize the `bcfishpass` access model, linear spawning/rearing habitat model and lateral habitat connectivity for planning purposes. These models provide a valuable starting point, but their results are not definitive and should always be considered with professional judgment. Detailed information regarding model methodology, select parameters and known model limitations are detailed in Norris ([2020] 2024) with key documentation linked below:

- [Access model](#)
- [Linear spawning/rearing habitat models](#)
- [Lateral habitat model](#)

Table 3.1 documents the custom species-specific thresholds for stream gradient and channel width applied to the linear spawning and rearing habitat model for this year’s project planning.

Table 3.1: Stream gradient and channel width thresholds used to model potentially highest value fish habitat.

Variable	Bull Trout	Arctic Grayling
Spawning Gradient Max (%)	5.5	2.5
Spawning Width Min (m)	2	4
Rearing Width Min (m)	1.5	1.5
Rearing Gradient Max (%)	10.5	3.5
*		

3.2.1.1 Statistical Support for `bcfishpass` Fish Habitat Modelling Updates

This project provided the statistical background for updates to `bcfishpass` that facilitated incorporation of channel width (observed or predicted) into species specific linear spawning/rearing habitat models. In early 2021, Bayesian statistical methods were developed to predict channel width in all provincial freshwater atlas stream segments where width measurements had not previously been measured in the field. The model was based on the relationship between watershed area and mean annual precipitation weighted by upstream watershed area (J. Thorley and Irvine 2021). In December of 2021, J. Thorley and Irvine (2021) methods were updated using a power model derived by Finnegan et al. (2005) which relates stream discharge to watershed area and mean annual precipitation resulting in J. L. Thorley, Norris, and Irvine (2021) which was utilized for

3.2 Planning

channel width estimates within `bcfishpass` modelling at the time of reporting. More detailed documentation of the methodology used to facilitate both the data collection and statistical analysis can be sourced in Irvine ([2021] 2022) and J. L. Thorley, Norris, and Irvine (2021).

In 2024, in collaboration with Poisson Consulting - stream discharge and temperature causal effects pathways were mapped with the intent of focusing aquatic restoration actions in areas of highest potential for positive impacts on fisheries values (ie. elimination of areas from intrinsic models where water temperatures are likely too cold to support fish production). The project began with a custom mechanistic model (visually represented [here](#)), but the model struggled to converge. The project then shifted to the `air2stream` model, which offers a middle ground between fully mechanistic models—often data-intensive and reliant on quantities that are difficult to measure or estimate—and purely statistical models, which lack physical justification and perform poorly when extrapolated to new conditions (Toffolon and Piccolroaz (2015)). After several adaptations, the expected stream temperatures were best modeled using the four-parameter version of the `air2stream` model, with added random effects by site for each of the four parameters (Hill, Thorley, and Irvine (2024)). The data used for the model were sourced from the following locations, for years 2019-2021:

- Water temperature data collected in the Nechako Watershed were downloaded from [Zenodo](#) (Gilbert et al. 2022).
- Hourly air temperature data were obtained from the ERA-5-Land dataset via the Copernicus Climate Change Service (Muñoz Sabater (2019))
- Daily baseflow and surface runoff data were sourced from the [Pacific Climate Impacts Consortium's Gridded Hydrologic Model Output](#) using the ACCESS1-0_rcp85 scenario (Pacific Climate Impacts Consortium (n.d.)).

3.2.2 Bull Trout and Arctic Grayling Critical Habitat

We collaborated with Eduardo Martins (Associate Professor at UNBC) to incorporate Arctic grayling into our collaborative GIS project so it could be used to inform field work activities and potential remediation site prioritization. Similarly, critical bull trout habitat data was added to the GIS project after being obtained from Luc Turcotte (WLRS - Regional Aquatic Specialist) and stored securely. Bull trout data acquisition required prior completion of the Access to Secure Species and Ecosystem Data and Information (SEDIS) training conducted through WLRS.

3.3 Fish Passage Assessments

3.3.1 Natural Barriers to Fish Passage

Our assessments may include natural features such as waterfalls that could limit fish passage. This informs whether upstream culvert upgrades would restore access for anadromous species (e.g., salmon) or primarily benefit resident fish already upstream. We document these features by measuring height, gradient, and pool depth, recording field observations, capturing site photographs, and reviewing background sources for context.

3.3.2 Road Stream Crossings

In the field, crossings prioritized for follow-up were first assessed for fish passage following the procedures outlined in “Field Assessment for Determining Fish Passage Status of Closed Bottomed Structures” (MoE 2011). The reader is referred to (MoE 2011) for detailed methodology. Crossings surveyed included closed bottom structures (CBS), open bottom structures (OBS) and crossings considered “other” (i.e. fords). Photos were taken at surveyed crossings and when possible included images of the road, crossing inlet, crossing outlet, crossing barrel, channel downstream and channel upstream of the crossing and any other relevant features. The following information was recorded for all surveyed crossings: date of inspection, crossing reference, crew member initials, Universal Transverse Mercator (UTM) coordinates, stream name, road name and kilometer, road tenure information, crossing type, crossing subtype, culvert diameter or span for OBS, culvert length or width for CBS. A more detailed “full assessment” was completed for all closed bottom structures and included the following parameters: presence/absence of continuous culvert embedment (yes/no), average depth of embedment, whether or not the culvert bed resembled the native stream bed, presence of and percentage backwatering, road fill depth, outlet drop, outlet pool depth, inlet drop, culvert slope, average downstream channel width, stream slope, presence/absence of beaver activity, presence/absence of fish at time of survey, type of valley fill, and a habitat value rating. Habitat value ratings were based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), fish migration patterns, the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation (Table 3.2).

Table 3.2: Habitat value criteria (Fish Passage Technical Working Group, 2011).

Habitat Value	Fish Habitat Criteria
High	The presence of high value spawning or rearing habitat (e.g., locations with abundance of suitably sized gravels, deep pools, undercut banks, or stable debris) which are critical to the fish population.
Medium	Important migration corridor. Presence of suitable spawning habitat. Habitat with moderate rearing potential for the fish species present.
Low	No suitable spawning habitat, and habitat with low rearing potential (e.g., locations without deep pools, undercut banks, or stable debris, and with little or no suitably sized spawning gravels for the fish species present).

3.3 Fish Passage Assessments

Fish passage potential was determined for each stream crossing identified as a closed bottom structure as per MoE (2011). The combined scores from five criteria: depth and degree to which the structure is embedded, outlet drop, stream width ratio, culvert slope, and culvert length were used to screen whether each culvert was a likely barrier to some fish species and life stages (Table 3.3, Table 3.4). These criteria were developed based on data obtained from various studies and reflect an estimation for the passage of a juvenile salmon or small resident rainbow trout (Clarkin et al. 2005 ; Bell 1991; Thompson 2013). For crossings determined to be potential barriers or barriers based on the data, a culvert fix and recommended diameter/span was proposed.

Table 3.3: Fish Barrier Risk Assessment (MoE 2011).

Risk	LOW	MOD	HIGH
Embedded	>30cm or >20% of diameter and continuous	<30cm or 20% of diameter but continuous	No embedment or discontinuous
Value	0	5	10
Outlet Drop (cm)	<15	15-30	>30
Value	0	5	10
SWR	<1.0	1.0-1.3	>1.3
Value	0	3	6
Slope (%)	<1	1-3	>3
Value	0	5	10
Length (m)	<15	15-30	>30
Value	0	3	6

Table 3.4: Fish Barrier Scoring Results (MoE 2011).

Cumulative Score	Result
0-14	passable
15-19	potential barrier
>20	barrier

The habitat gain index is the quantity of modelled habitat upstream of the subject crossing and represents an estimate of habitat gained with remediation of fish passage at the crossing. For this project, a gradient threshold between accessible and non-accessible habitat was set at 25% (for a minimum length of 100m) intended to represent the maximum gradient of which the strongest swimmers of anadromous species (bull trout) are likely to be able to migrate upstream.

3 Methods

For reporting of Phase 1 - fish passage assessments within the body of this report (Table 3.3), a “total” value of habitat <20% output from *bcfishpass* was used to estimate the amount of habitat upstream of each crossing less than 25% gradient before a falls of height >5m - as recorded in MoE (2020) or documented in other *bcfishpass* online documentation. For Phase 2 - habitat confirmation sites, conservative estimates of the linear quantity of habitat to be potentially gained by fish passage restoration, bull trout rearing maximum gradient threshold (10.5%) was used. To generate estimates for area of habitat upstream (m^2), the estimated linear length was multiplied by half the downstream channel width measured (overall triangular channel shape) as part of the fish passage assessment protocol. Although these estimates are not generally conservative, have low accuracy and do not account for upstream stream crossing structures they allow a rough idea of the best candidates for follow up.

Potential options to remediate fish passage were selected from MoE (2011) and included:

- Removal (RM) - Complete removal of the structure and deactivation of the road.
- Open Bottom Structure (OBS) - Replacement of the culvert with a bridge or other open bottom structure. Based on consultation with FLNR road crossing engineering experts, for this project we considered bridges as the only viable option for OBS type .
- Streambed Simulation (SS) - Replacement of the structure with a streambed simulation design culvert. Often achieved by embedding the culvert by 40% or more. Based on consultation with FLNR engineering experts, we considered crossings on streams with a channel width of <2m and a stream gradient of <8% as candidates for replacement with streambed simulations.
- Additional Substrate Material (EM) - Add additional substrate to the culvert and/or downstream weir to embed culvert and reduce overall velocity/turbulence. This option was considered only when outlet drop = 0, culvert slope <1.0% and stream width ratio < 1.0.
- Backwater (BW) - Backwatering of the structure to reduce velocity and turbulence. This option was considered only when outlet drop < 0.3m, culvert slope <2.0%, stream width ratio < 1.2 and stream profiling indicates it would be effective..

3.3.3 Cost Estimates

Cost estimates for structure replacement with bridges and embedded culverts were generated based on the channel width, slope of the culvert, depth of fill, road class and road surface type. Road details were sourced from FLNRORD (2020b) and FLNRORD (2020a) through *bcfishpass*. Interviews with Phil MacDonald, Engineering Specialist FLNR - Kootenay, Steve Page, Area Engineer - FLNR - Northern Engineering Group and Matt Hawkins - MoTi - Design Supervisor for Highway Design and Survey - Nelson were utilized to help refine estimates which have since been adjusted for inflation in 2020 and based on past experience.

Base costs for installation of bridges on forest service roads and permit roads with surfaces specified in provincial GIS road layers as rough and loose was estimated at \$30000/linear m and

3.3 Fish Passage Assessments

assumed that the road could be closed during construction and a minimum bridge span of 15m. For streams with channel widths <2m, embedded culverts were reported as an effective solution with total installation costs estimated at \$100k/crossing (pers. comm. Phil MacDonald, Steve Page then adjusted for inflation in 2020). For larger streams (>6m), estimated span width increased proportionally to the size of the stream. For crossings with large amounts of fill (>3m), the replacement bridge span was increased by an additional 3m for each 1m of fill >3m to account for cutslopes to the stream at a 1.5:1 ratio. To account for road type, a multiplier table was generated to estimate incremental cost increases with costs estimated for structure replacement on paved surfaces, railways and arterial/highways costing up to 15 times more than forest service roads due to expenses associate with design/engineering requirements, traffic control and paving. The cost multiplier table (Table 3.5) should be considered very approximate with refinement recommended for future projects.

Table 3.5: Cost multiplier table based on road class and surface type.

Class	Surface	Class Multiplier	Surface Multiplier	Bridge \$/15m	Streambed Simulation \$
FSR	Rough	1	1	450,000	100,000
FSR	Loose	1	1	450,000	100,000
Resource	Loose	1	1	450,000	100,000
Collector	Loose	1	1	450,000	100,000
Recreation	Loose	1	1	450,000	100,000
Resource	Rough	1	1	450,000	100,000
Permit	Unknown	1	1	450,000	100,000
Permit	Loose	1	1	450,000	100,000
Permit	Rough	1	1	450,000	100,000
Unclassified	Loose	1	1	450,000	100,000
Unclassified	Rough	1	1	450,000	100,000
Unclassified	Paved	1	2	750,000	150,000
Unclassified	Unknown	1	2	750,000	150,000
Local	Loose	4	1	1,500,000	200,000
Local	Paved	4	2	3,000,000	400,000
Collector	Paved	4	2	3,000,000	400,000
Arterial	Paved	15	2	11,250,000	1,500,000
Highway	Paved	15	2	11,250,000	1,500,000
Rail	Rail	15	2	11,250,000	1,500,000

3.3.4 Climate Change Risk Assessment

In collaboration with the Ministry of Transportation and Infrastructure (MoTi), a new climate change replacement program aims to prioritize vulnerable culverts for replacement (pers. comm Sean Wong, 2022) based on data collected and ranked related to three categories - culvert condition, vulnerability and priority. Within the “condition” risk category - data was collected and crossings were ranked based on erosion, embankment and blockage issues. The “climate” risk category included ranked assessments of the likelihood of both a flood event affecting the culvert as well as the consequence of a flood event affecting the culvert. Within the “priority” category the following factors were ranked - traffic volume, community access, cost, constructability, fish bearing status and environmental impacts (Table 3.6). This project is still in its early stages with methodology changes going forward.

Table 3.6: Climate change data collected at MoTi culvert sites

Parameter	Description
erosion_issues	Erosion (scale 1 low - 5 high)
embankment_fill_issues	Embankment fill issues 1 (low) 2 (medium) 3 (high)
blockage_issues	Blockage Issues 1 (0-30%) 2 (>30-75%) 3 (>75%)
condition_rank	Condition Rank = embankment + blockage + erosion
condition_notes	Describe details and rational for condition rankings
likelihood_flood_event_affecting_culvert	Likelihood Flood Event Affecting Culvert (scale 1 low - 5 high)
consequence_flood_event_affecting_culvert	Consequence Flood Event Affecting Culvert (scale 1 low - 5 high)
climate_change_flood_risk	Climate Change Flood Risk (likelihood x consequence) 1-6 (low) 6-12 (medium) 10-25 (high)
vulnerability_rank	Vulnerability Rank = Condition Rank + Climate Rank
climate_notes	Describe details and rational for climate risk rankings
traffic_volume	Traffic Volume 1 (low) 5 (medium) 10 (high)
community_access	Community Access - Scale - 1 (high - multiple road access) 5 (medium - some road access) 10 (low - one road access)
cost	Cost (scale: 1 high - 10 low)
constructability	Constructability (scale: 1 difficult -10 easy)
fish_bearing	Fish Bearing 10 (Yes) 0 (No) - see maps for fish points
environmental_impacts	Environmental Impacts (scale: 1 high -10 low)
priority_rank	Priority Rank = traffic volume + community access + cost + constructability + fish bearing + environmental impacts
overall_rank	Overall Rank = Vulnerability Rank + Priority Rank
priority_notes	Describe details and rational for priority rankings

3.4 Habitat Confirmation Assessments

Following fish passage assessments, habitat confirmations were completed in accordance with procedures outlined in the document “A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation of a Stream Crossing” (Fish Passage Technical Working Group 2011). The main objective of the field surveys was to document upstream habitat quantity and quality and to determine if any other obstructions exist above or below the crossing. Habitat value was assessed based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation. Criteria used to rank habitat value was based on guidelines in Fish Passage Technical Working Group (2011) (Table [3.2](#)).

During habitat confirmations, to standardize data collected and facilitate submission of the data to provincial databases, information was collected on digital field forms adapted from provincial [“Site Cards”](#). Habitat characteristics recorded included channel widths, wetted widths, residual pool depths, gradients, bankfull depths, stage, temperature, conductivity, pH, cover by type, substrate and channel morphology (among others). When possible, the crew surveyed downstream of the crossing to a minimum distance 300m and upstream to a minimum distance of 500 - 600m. Any potential obstacles to fish passage were inventoried with photos, physical descriptions and locations recorded on site cards. Surveyed routes were recorded with time-signatures on handheld GPS units.

3.4.1 Fish Sampling

3.4.1.1 Electrofishing

Fish sampling was conducted on a subset of sites when biological data was considered to add significant value to the physical habitat assessment information. Electrofishing was utilized for fish sampling according to stream inventory standards and procedures found in the Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Manual (Resources Inventory Committee 2001). A Haltech 2000 backpack electrofisher was used within discrete site units both upstream and downstream of the subject crossing with electrofisher settings and seconds, water quality parameters (i.e. conductivity, temperature and pH), start and end locations, length of site and wetted widths (average of a minimum of three) recorded.

3.4.1.2 Fish Handling and Processing

Captured fish were held in buckets with sufficient water to minimize stress until processing, and multiple buckets were used when catch numbers were high. For each fish captured, fork length, weight and species was recorded with results documented in the fish data submission spreadsheet.

3.4.1.3 Pit Tagging

Fish with a fork length greater than 60 mm and belonging to species approved under the scientific fish collection permit PG25-983916 were tagged with Passive Integrated Transponders (PIT tags) using the [Abdominal Cavity](#) method outlined by Biomark. To anesthetize fish prior to pit tagging, we used a solution of approximately 0.1 mL of clove oil per 1 L of water (1:10,000). This concentration was selected for its efficiency in providing effective sedation with minimal residual effects, making it ideal for studies in which fish are released back into their natural habitats (Fernandes et al. 2017). The clove oil solution was prepared in advance by dissolving pure clove oil in ethyl alcohol in a 1:9 ratio (clove oil: ethyl alcohol) to enhance solubility, then mixed into the water bucket (Fernandes et al. 2017). Fish were immersed in this solution until they reached an appropriate level of anesthesia for handling and then were tagged. To maintain needle sharpness and minimize injury risk, needles were replaced approximately every 10 fish. Each tagged fish was scanned with the PIT reader, and both the PIT tag ID and row ID were recorded. Once tagged, fish were placed into a bucket of fresh water and allowed to recover before being released back into the stream. Fish information and habitat data will be submitted to the province under scientific fish collection permit PG25-983916.

3.4.2 Environmental DNA (eDNA) Sampling

3.4.2.1 Sample Collection and Filtration

Samples were collected using two approaches informed by Hobbs, Helbing, and Goldberg (2025): in-stream filtration with a Haltech EMOS sampler and grab-and-go bottle collection. For both methods, all water collected at a site (typically 1–3L) was filtered onto a single filter, yielding one replicate per site. Mixed cellulose ester (MCE) filters with 1 μ m pore size were used most commonly, with 0.45 μ m and 1.5 μ m filters of other materials used occasionally. Samples were typically collected upstream and downstream of each road–stream crossing, though in some cases only downstream samples were obtained.

For in-stream filtration, Haltech reusable aluminum housings were pre-loaded with filters. In high-turbidity conditions, a Haltech pre-filter with a 100 μ m nitex screen was installed upstream to remove coarse material. For grab-and-go sampling, stream water was collected in sanitized 1L HDPE Nalgene bottles and filtered at the end of each day.

After filtration, filters were removed from housings in a clean environment, placed in coin envelopes with silica desiccant inside labelled ziplock bags, and frozen until shipment. Aluminum housings and Nalgene bottles were sanitized with bleach solution between uses.

3.4.2.2 Quality Assurance

Approximately 10% of samples were collected for quality assurance. Field blanks using distilled water were collected daily. Office blanks were processed at field accommodations by filtering clean water when processing grab-and-go samples. Positive controls were obtained at select sites where electrofishing confirmed species presence prior to eDNA sampling.

3.4.2.3 Laboratory Analysis

Samples were shipped frozen to the University of Northern British Columbia (UNBC) for extraction and analysis. DNA was extracted and analyzed using quantitative polymerase chain reaction (qPCR) with species-specific assays following protocols developed by Murray and Booth (2023). Assays targeted mitochondrial genes (COI, CytB) using fluorescent dyes (FAM, VIC, HEX), allowing multiple species to be tested per sample through multiplexing. Target species included Chinook, Coho, Sockeye, Rainbow Trout, Bull Trout, Arctic Grayling, and Kokanee, with additional assays for Coastal Cutthroat and White Sturgeon at select sites.

Due to close evolutionary relationships and hybridization, some species pairs cannot be distinguished using mtDNA-based assays. The Bull Trout assay also detects Dolly Varden, requiring geographic context to interpret results in areas where ranges overlap. Similarly, Rainbow Trout and Coastal Cutthroat assays may cross-react where these species co-occur or hybridize.

To optimize laboratory efficiency, we collaborated with UNBC to develop a sample run plan that grouped samples by compatible dye types, minimizing the number of qPCR runs required. An [assay mapping table](#) cross-referencing species codes to assay identifiers and dye types was developed, along with [processing scripts](#) to generate sample groupings and a [run order configuration](#) specifying species order within each group.

3.4.2.4 eDNA Extraction

eDNA was extracted in the dedicated eDNA lab at UNBC and analysed with the protocols developed by Murray and Booth (2023).

3.4.3 Open Source Reporting Framework

This report is produced using open-source tools (R, bookdown) with full version control via git, enabling iterative updates. All code, data, and revision history are publicly accessible in the [project repository](#), with ongoing tasks tracked via GitHub [issues](#) and version history maintained in the [project changelog](#).

3.4.4 Aerial Imagery

Scripted processing and serving of UAV imagery collected during the project is available at https://github.com/NewGraphEnvironment/stac_uav_bc/ (Irvine [2025] 2025). [OpenDroneMap](#) was utilized to produce orthomosaics, digital surface models (DSMs), and digital terrain models (DTMs) (OpenDroneMap Authors [2014] 2025). To support efficient web-based access - imagery products were converted to cloud-optimized GeoTIFFs (COGs) using `rio-cogeo`, then collated according to the [SpatioTemporal Asset Catalog \(STAC\)](#) specification with `pystac` and uploaded to S3 storage Amazon Web Services (2025). A `titiler` tile server was set up to facilitate interactive viewing of the orthoimagery and an Application Program Interface (API) leveraging `stac-fastapi-pgstac` is served at <https://images.a11s.one> to enable linking of collection images through QGIS as well as remote spatial and temporal querying using open source software such as `rstac` (Development Seed [2019] 2025; `stac-utils` 2025; Simoes et al. 2021).

3.5 Engineering Design

Engineering designs were signed and sealed by professional engineers. When possible - completed designs are loaded to the PSCIS data portal.

3.6 Remediations

Structure replacement was conducted by project specific contractors. If not already completed, as-built drawings will be loaded to the PSCIS data portal.

3.7 Monitoring

Monitoring of fish passage restoration sites — both proposed and completed - is essential to ensure restoration investments lead to meaningful ecological outcomes. Monitoring enables evaluation of whether remediation actions improve connectivity for fish and provides critical feedback to refine future prioritization and restoration strategies.

Baseline data collection, including fish sampling and aerial surveys via drone, are core components of this monitoring. While detailed methods for these activities are included in the previous habitat confirmation section, they are also fundamental to effectiveness monitoring, as they provide context to not only facilitate prioritization and communications but also for detecting change following restoration.

To support consistent and targeted assessment, a custom field form was developed for routine effectiveness monitoring based loosely on Forest Investment Account (2003) but tailored specifically for fish passage projects. Table 3.7) outlines the monitoring metrics used.

Table 3.7: Description of monitoring metrics used for effectiveness monitoring.

Parameter	Description
Dewatering	Have the remediation works led to dewatering of the channel due to substrate aggradation or other factors?
Velocity	Are flow velocities similar to those within the natural channel? Are they expected to exceed swim speeds of particular fish species/life stages of interest?
Constriction	Have the remediation works led to constriction of the channel. Compare channel width underneath structure and within construction footprint to average channel widths upstream and downstream?
Substrate	Is the substrate within/under and adjacent to the remediated structure generally equivalent to that found upstream and downstream where natural channel conditions exist?
Riparian	What is the condition of the riparian area within the construction footprint?
UAV Flight	Was a flight conducted with unmanned aerial vehicle to document conditions at time of monitoring?
Flow_depth	What are the flow depths at the time of assessment within project footprint. Are depths expected to be sufficient to facilitate upstream passage for specific species/life stages of interest?

3.7 Monitoring

Parameter	Description
Stability	Does the structure appear to be stable or is there evidence of erosion/shifting?
Revegetation	How were riparian areas rehabilitated and are they improving fish habitat value?
Cover	Is cover available for fish within the construction footprint in the form of overhanging vegetation, large/small woody debris, boulders, undercut banks, etc?
Maintenance	If required, provide maintenance recommendations.
Recommendations	General recommendations for follow up. Could include revegetation, addition of substrate, fish sampling, etc.

4 Results and Discussion

4.1 Site Assessment Data

Fish passage assessment procedures conducted through SERNbc since 2020 are amalgamated online within the Results and Discussion section of the report found [here](#).

Since 2020, orthoimagery and elevation model rasters have been generated and stored as Cloud Optimized Geotiffs on a cloud service provider (AWS) with select imagery linked to in the collaborative GIS project. Additionally - a tile service has been set up to facilitate viewing and downloading of individual images, provided

[athttps://www.newgraphenvironment.com/fish_passage_peace_2025_reporting//results-and-discussion.html](https://www.newgraphenvironment.com/fish_passage_peace_2025_reporting//results-and-discussion.html).

Table 4.1 :
Summary
of fish
passage
assessment
procedures
conducted
in northern
British
Columbia
through
SERNbc.

**NOTE: To
view all
columns in
the table -
please
click on
one of the
sort
arrows
within
column
headers
before
scrolling
to the
right.**

4 Results and Discussion

Show 5 entries

Watershed Group	Assessment	Reassessment	Habitat Confirmation	Design	Rem
All	All	All	All	All	All
Francois Lake	1	0	4	0	0
Lower Chilako River	1	0	1	0	0
Morkill River	1	0	5	0	0
Nechako River	2	0	4	0	0
Tabor River	0	1	0	0	0
Upper Fraser River	2	2	3	0	0
Total	7	3	17	0	0

Showing 1 to 7 of 7 entries

Excel

CSV

Previous

1

Next

Table 4.2:
Details of
fish
passage
assessment
procedures
conducted
in northern
British
Columbia
through
SERNbc.

Show 5 entries

4.1 Site Assessment Data

Stream Crossing Id	My Crossing Reference	Watershed Group	Stream Name	Road Name	Assessment
All	All	All	All	All	All
4931		Upper Fraser River	Crooked Creek	Tinsley Pit Road	yes
7620		Upper Fraser River	Tributary of McLennan River	CN Rail access road	yes
7622		Francois Lake	UNN Flows into Fraser Lake	Stella Rd	
196200		Tabor River	Bittner Creek	Forman Road	yes
199171		Francois Lake	Tributary to Fraser Lake	Gala Bay Road	
199172		Francois Lake	Scotch Creek	Stella Road	
199173		Nechako River	Tributary to Nechako River	Dog Creek Road	
199174		Nechako River	Tributary to Nechako River	Sutherland FSR	
199190		Nechako River	Clear Creek	Highway 27 S	
199204		Nechako River	Nine Mile Creek	Settlement Rd	

Showing 1 to 10 of 26 entries

[Excel](#)
[CSV](#)
[Previous](#)
[1](#)
[2](#)
[3](#)
[Next](#)

4.2 Collaborative GIS Environment

In addition to numerous layers documenting fieldwork activities since , a summary of background information spatial layers and tables loaded to the collaborative GIS project (sern_peace_fwcp_2023) at the time of writing (2026-05-04) are included online [here](#)

4.3 Planning

4.3.1 Habitat Modelling

Habitat modelling from `bcfishpass` including access model, linear spawning/rearing habitat model and lateral habitat connectivity models for watershed groups within our study area were updated for the spring of 2025 and are included spatially in the collaborative GIS project. A snapshot of these outputs related to each modeled and PSCIS stream crossing structure are also included within an `sqlite` database within this year's project reporting/code repository [here](#).

4.3.1.1 Statistical Support for `bcfishpass` Fish Habitat Modelling Updates

Initial mapping of stream discharge and temperature causal effects pathways for the future purpose of focusing aquatic restoration actions in areas of highest potential for positive impacts on fisheries values (ie. elimination of areas from intrinsic models where water temperatures are likely too cold to support fish production) are detailed in Hill, Thorley, and Irvine (2024) which is included as [Attachment - Water Temperature Modelling \(page 99\)](#).

4.4 Fish Passage Assessemnts

Field assessments were conducted between September 21, 2025 and September 23, 2025 by Allan Irvine, R.P.Bio. and Lucy Schick, B.Sc. A total of 16 Fish Passage Assessments were completed, including 15 Phase 1 assessments and 1 reassessments.

Of the 16 sites where fish passage assessments were completed, 15 were not yet inventoried in the PSCIS system. This included 3 crossings considered “passable”, 1 crossings considered a “potential” barrier, and 11 crossings were considered “barriers” according to threshold values based on culvert embedment, outlet drop, slope, diameter (relative to channel size) and length (MoE 2011).

A summary of crossings assessed, a rough cost estimate for remediation, and a priority ranking for follow-up for Phase 1 sites is presented in Table [4.3](#). Detailed data with photos are presented in [Attachment - Phase 1 Data and Photos \(page 95\)](#).

4.4 Fish Passage Assessemnts

The “Barrier” and “Potential Barrier” rankings used in this project followed MoE (2011) and represent an assessment of passability for juvenile salmon or small resident rainbow trout under any flow conditions that may occur throughout the year (Clarkin et al. 2005; Bell 1991; Thompson 2013). As noted in Bourne et al. (2011), with a detailed review of different criteria in Kemp and O’Hanley (2010), passability of barriers can be quantified in many different ways. Fish physiology (i.e. species, length, swim speeds) can make defining passability complex but with important implications for evaluating connectivity and prioritizing remediation candidates (Bourne et al. 2011; Shaw et al. 2016; Mahlum et al. 2014; Kemp and O’Hanley 2010). Washington Department of Fish & Wildlife (2009) present criteria for assigning passability scores to culverts that have already been assessed as barriers in coarser level assessments. These passability scores provide additional information to feed into decision making processes related to the prioritization of remediation site candidates and have potential for application in British Columbia.

4 Results and Discussion

Table 4.3: Upstream habitat estimates and cost benefit analysis for Phase 1 assessments ranked as a ‘barrier’ or ‘potential’ barrier. Bull trout network model used for habitat estimates (total length of stream network <25% gradient).

PSCIS ID	External ID	Priority	Stream	Road	Barrier Result	Habitat value	Habitat Upstream (km)	Stream Width (m)	Fix	Cost Est (\$K)
6559	–	low	Tributary To Williston Reservoir	Finlay Community Connector	Barrier	Low	17.4	1.0	SS-CBS	100
203597	15201563	high	Tributary To Nation River	Thut 15000	Barrier	High	16.8	2.5	OBS	450
203598	15201146	moderate	Tributary To Nation River	Thut 15000	Barrier	Medium	9.2	1.6	SS-CBS	100
203599	15202249	moderate	Tributary To Nation River	Thut 14000	Barrier	Medium	14.2	3.0	OBS	450
203600	15200985	moderate	Tributary To Nation River	Finlay-Nation FSR	Barrier	High	30.6	4.1	OBS	450
203601	15201007	moderate	Tributary To Nation River	Finlay-Nation FSR	Barrier	High	15.9	3.5	OBS	450
203602	15201006	high	Tributary To Nation River	Finlay-Nation FSR	Barrier	High	17.5	3.4	OBS	495
203603	16400714	moderate	Tributary To Williston Reservoir	Finlay-Nation FSR	Barrier	High	106.7	7.4	OBS	450
203604	16401990	moderate	Tributary To Williston Reservoir	Finlay FSR	Barrier	Medium	26.6	10.0	OBS	810

4.5 Habitat Confirmation Assessments

During the 2025 field assessments, habitat confirmation assessments were conducted at 3 sites within the Carp Lake, Crooked River, Nation River, Parsnip Arm, and Parsnip River watershed groups. A total of approximately 2km of stream was assessed. Georeferenced field maps are provided in [Attachment - Maps \(page 93\)](#).

As collaborative decision-making was ongoing at the time of reporting, site prioritization can be considered preliminary. Results are summarized in Table [4.4](#) with raw habitat and fish sampling data included in [Attachment - Data \(page 97\)](#). A summary of preliminary modelling results illustrating quantities of bull trout spawning and rearing habitat potentially available upstream of each crossing as estimated by measured/modelled channel width and upstream accessible stream length are presented in Figure [4.1](#). Detailed information for each site assessed with Phase 2 assessments (including maps) are presented within site-specific appendices to this document.

4.5 Habitat Confirmation Assessments

Table 4.4: Overview of habitat confirmation sites. Bull trout rearing model used for habitat estimates (total length of stream network <10.5% gradient).

PSCIS ID	Stream	Road	Tenure	UTM	UTM zone	Fish Species	Habitat Gain (km)	Habitat Value	Priority	Comments
199663	Tributary To Parsnip River	Chco 12000	BCTS	515416 6094963	10	–	0.0	Medium	Moderate	Low gradient stream with significant riparian tree blow down throughout. Very difficult walking so most was done outside of the active channel with spot visits to the stream for measurements/photos completed intermittently for the 600m that was assessed. Wide floodplain within well defined canyon (approximately 15m deep with slopes of ~45°). Healthy riparian, consisting of herb and shrub cover within sparse spruce forest that appears to have been impacted by some sort of beetle or other affliction. Most of the spruce trees were dead with many already fallen down.
203597	Tributary To Nation River	Thut 15000	Sekani R07214 8	425959 6122703	10	–	5.6	High	High	Surveyed 350m upstream to the beginning of the wetlands. Very nice stream with good flow, although recent rainfall may have had an influence. Pockets of gravels throughout the area surveyed. Although the area viewed was limited, the wetlands appeared shallow so freezing to the bottom may occur during the winter.
203605	Tributary To Williston Reservoir	Finlay FSR	MoF	486504 6124990	10	–	35.9	High	High	Large stream with complex habitat and diverse cover including deep pools, boulders, large and small woody debris, and undercut banks. A canyon section approximately 100m long was located 100m upstream of the Modeste FSR bridge, containing multiple cascade steps up to 1m high with deep downstream pools. At the upper end of the site, the channel deflected against steep gully walls. Abundant gravel was present, suitable for adfluvial and resident rainbow trout as well as bull trout. eDNA sample collected.

Table 4.5: Summary of Phase 2 fish passage reassessments.

PSCIS ID	Embedded	Outlet Drop (m)	Diameter (m)	SWR	Slope (%)	Length (m)	Final score	Barrier Result
199663	No	0.60	1.2	0.00	3	13	30	Barrier
203597	No	0.25	0.8	3.12	1	19	29	Barrier

4 Results and Discussion

PSCIS ID	Embedded	Outlet Drop (m)	Diameter (m)	SWR	Slope (%)	Length (m)	Final score	Barrier Result
203605	No	0.95	3.9	1.54	2	46	37	Barrier

4.5 Habitat Confirmation Assessments

Table 4.6: Cost benefit analysis for Phase 2 assessments. Bull trout rearing model used for habitat estimates (total length of stream network <10.5% gradient).

PSCIS ID	Stream	Road	Barrier Result	Habitat value	Stream Width (m)	Fix	Cost Est (\$K)	Habitat Upstream (m)	Cost Benefit (m / \$K)	Cost Benefit (m2 / \$K)
199663	Tributary To Parsnip River	Chco 12000	Barrier	Medium	3.0	SS-CBS	–	0	–	–
203597	Tributary To Nation River	Thut 15000	Barrier	High	2.5	OBS	450	5570	12377.8	15472.2
203605	Tributary To Williston Reservoir	Finlay FSR	Barrier	High	6.7	OBS	450	35932	79848.9	239546.7

4 Results and Discussion

Table 4.7: Summary of Phase 2 habitat confirmation details.

PSCIS ID	Length surveyed upstream (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
-	-	-	-	-	-	-	-
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4.5 Habitat Confirmation Assessments

Table 4.8: Summary of watershed area statistics upstream of Phase 2 crossings.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
199663	1.0	713	707	849	784	774	SW
203597	8.2	900	907	1065	980	971	S
203605	55.6	681	671	1644	980	937	SE
203611	55.6	682	671	1644	980	937	SE

* Elev P60 = Elevation at which 60% of the watershed area is above

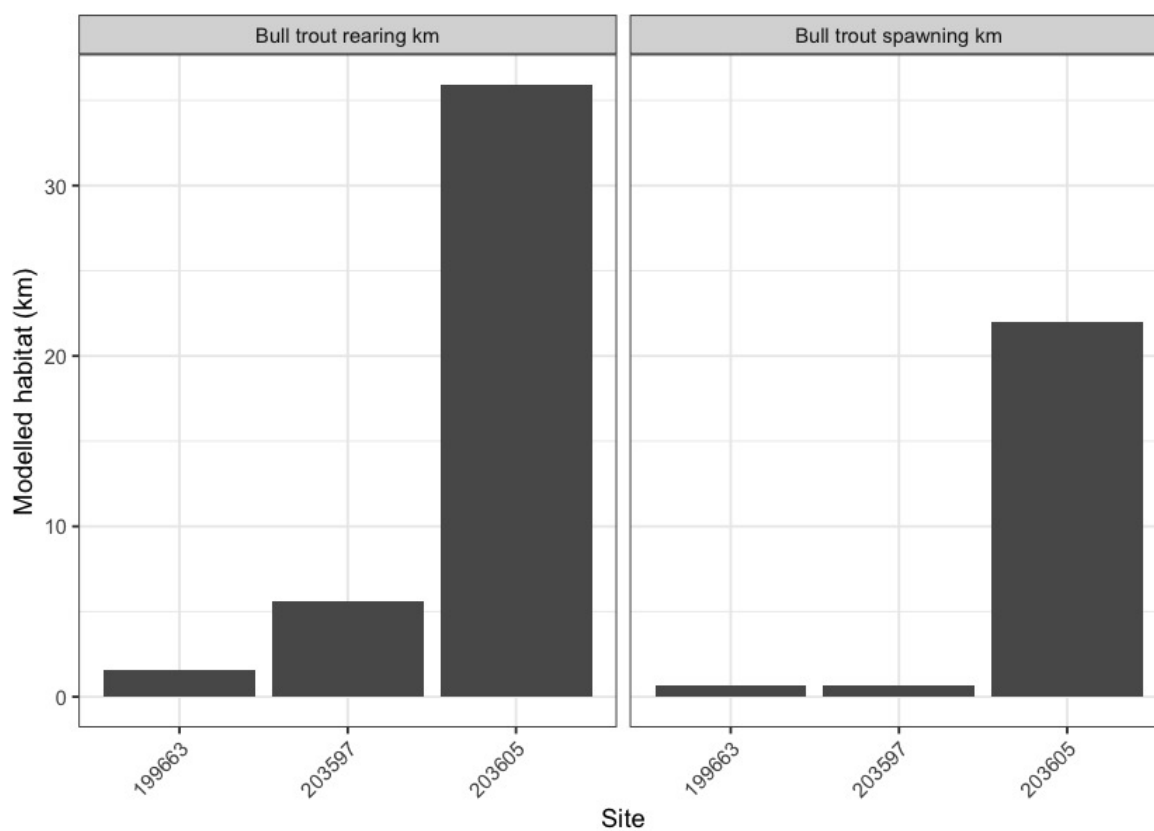


Figure 4.1: Summary of potential rearing and spawning habitat upstream of habitat confirmation assessment sites. Bull trout rearing and spawning models used for habitat estimates (total length of stream network <10.5% and <5.5% gradient, respectively).

4.5.1 Environmental DNA (eDNA) sampling

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 35 collection events were undertaken across 10 streams, including 28 real environmental samples, 3 field blanks, and 4 office blanks for quality assurance.

Per-species detection results across the 28 real environmental samples are summarized in Table 4.9. Detections are based on the standard ddPCR call threshold of ≥ 4 positive droplets; sub-threshold signals (1-3 droplets) are tracked separately as they are not interpretable as species presence per ddPCR convention. Site-level detail (per-site detection table, field blanks, and retests) is provided in [Appendix - Environmental DNA Results \(page 79\)](#), and the [interactive Peace eDNA map](#) shows toggleable per-species layers.

Table 4.9: Per-species summary of eDNA results across real environmental samples in the Peace project area (excludes field and office blanks).

Species	Code	Sites tested	Detected (≥ 4 droplets)	Sub-threshold only (1-3 droplets)	No detection (0 droplets)
Rainbow Trout	RAIN	28	22	2	4
Bull Trout	BULT	28	1	8	19
Burbot	BURB	2	1	0	1
Arctic Grayling	GRAY	24	0	1	23
Kokanee	SOCK	22	0	0	22

4.6 Engineering Design

MIGHT NEED TO UPDATE

No new designs were commissioned in due to uncertainty related to forest harvesting activities and lack of funding for the 50% costs of replacing structures. All sites with a design for remediation of fish passage can be found online within the Results and Discussion section of the report found [here](#).

4.7 Monitoring

In 2025, baseline or follow up monitoring data was gathered through completion of an effectiveness monitoring form (sites where remediation has been completed) and/or through baseline fish sampling and acquisition of aerial imagery at the following sites (Table 4.10:

- In 2025, effectiveness monitoring and eDNA sampling was conducted on Tributary to Table River (PSCIS crossing 125231) on Table FSR. The initial habitat confirmation for the site is

4.7 Monitoring

documented in Irvine (2020)[here](#). Baseline monitoring from 2023-2025 can be reviewed in [here](#).

- In 2025, effectiveness monitoring and eDNA sampling was conducted on Tributary to Missinka River (PSCIS crossing 125179) on Chuchinka-Missinka FSR. Background information regarding the site can be reviewed [here](#) with 2025 results added to this memo here [here](#).

Table 4.10: Summary of monitoring sites.

PSCIS ID	Stream	Road	Crossing type	Crossing subtype	Diameter or span (m)	Length or width (m)	Assessment comments
125231	Tributary to Table River	Table FSR	OBS	BRIDGE	–	–	Overall, the site looks very similar to last year, in good condition with no evidence of any structural issues and the instream portion doing quite well. As there is a large amount of rock rip wrap adjacent to the stream that was not planted the riparian vegetation within the footprint is basically nonexistent. Electrofishing was conducted at sites downstream and upstream of the bridge with fish over 65mm PIT tagged. eDNA samples were collected as well.
125179	Tributary to Missinka River	Chuchinka-Missinka FSR	OBS	BRIDGE	16	4.8	Routine effectiveness evaluation. Structure was replaced several years ago and monitoring is ongoing. Electrofishing with fish over 65mm in length PIT tagged, as well as EDNA sampling was conducted in 2025.

5 Recommendations

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices—picking battles carefully, building trust, and staying committed to a longer-term transformation.

While preliminary top remediation priorities are provided by watershed group, these rankings are inherently subjective and can depend on the capacity and willingness of infrastructure owners and tenure holders to support implementation—both financially and over the often multi-year project timelines. In practice, we must often act opportunistically, pursuing simpler, lower-cost options to maintain momentum and achieve near-term progress.

Government, community groups, landowners, non-profits, industry and other stakeholders should work collaboratively to address high and moderate priority barriers identified online within the Results and Discussion section of the report found [here](#). Although the table presents many options - currently - linked reports specify whether each site is a low, moderate, or high priority. Progress on any front is meaningful, and aiming to remediate at least one high-priority site per year per watershed group—regardless of its overall rank—is a practical and effective approach.

To enhance fish passage restoration in the FWCP Peace Region:

- Maintain strong partnerships to support funding, site selection, remediation, and monitoring through adaptive management informed by traditional knowledge and real-time data.
- Prioritize detailed assessments in areas with blockages and high habitat potential, especially near McLeod Lake.
- Use climate modeling to prioritize crossings that enable access to cold, drought-resistant habitats.
- Secure financial commitments for Fern Creek remediation despite uncertainties in harvest planning.

5 Recommendations

- Continue effectiveness monitoring at key sites using fish sampling, eDNA, PIT tagging, temperature data, and aerial imagery.
- Continue to develop a cost-effective monitoring framework to assess productivity gains from improved passage.
- Collaborate with WLRS, UNC, local fisheries experts, FWCP, and the CEMPRA Project working group.
- Utilize environmental DNA (eDNA) to better understand bull trout and Arctic grayling habitat use at both potential and remediated sites. This will refine prioritization and assess fish passage effectiveness.

Tributary To Parsnip River - 199663 - Appendix

Site Location

COST ESTIMATEs, CORRECT BCFISHPASS HABITAT MODELLING, and memo map (needs to be updated) SHOULD WORK ONCE THIS IS FIXED https://github.com/NewGraphEnvironment/fish_passage_template_reporting/issues/142

PSCIS crossing 199663 is located on Tributary To Parsnip River, approximately 20km northeast of community of McLeod Lake, BC, in the Parsnip River watershed group (Figure 5.1). The crossing is located 780m upstream of the stream's confluence with the Parsnip River on Chco 12000 road which is the responsibility of BC Timber Sales.

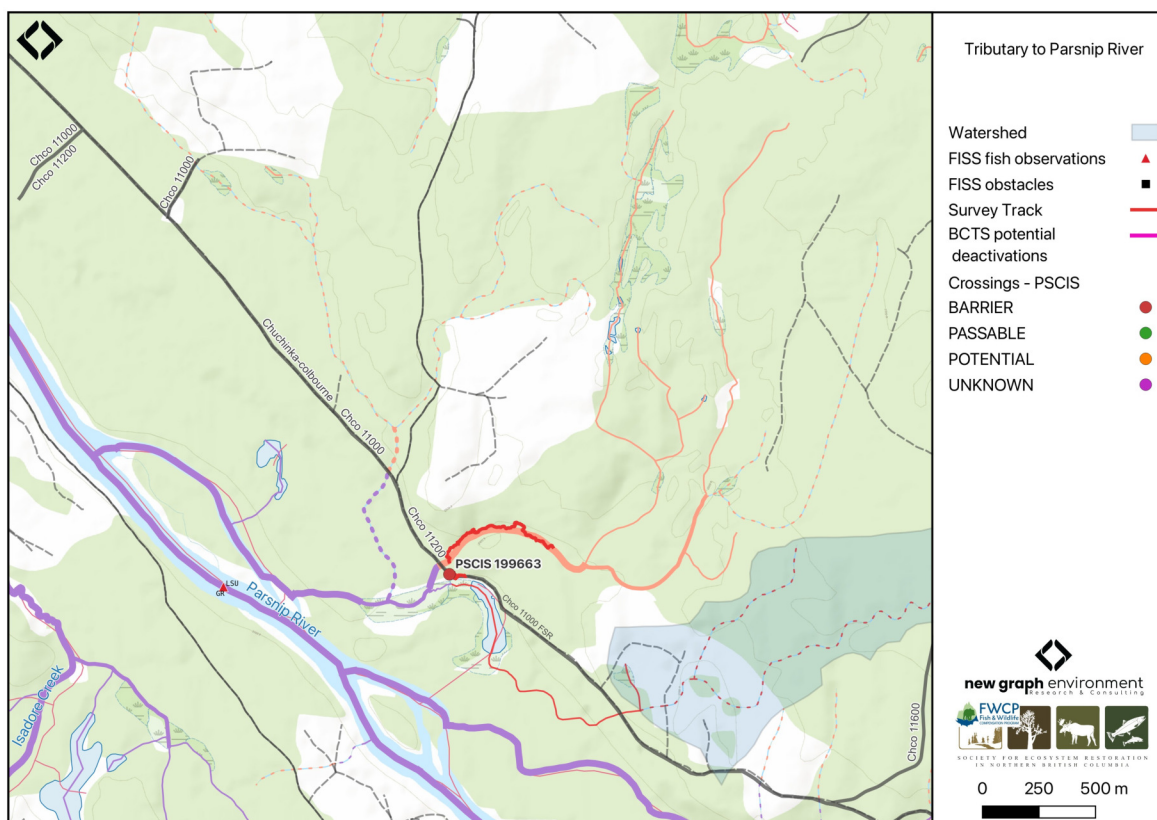


Figure 5.1: Map of Tributary To Parsnip River

Background

At PSCIS crossing 199663, Tributary To Parsnip River is a third order stream and drains a watershed of approximately 1km². The watershed ranges in elevation from a maximum of 849m to

713m near the crossing (Table [5.1](#)).

PSCIS crossing 199663 was first assessed in 2024 as part of this project and was ranked as a barrier to fish passage with medium value habitat documented (Irvine and Schick 2025). The crossing was reassessed in 2025, and a habitat confirmation was completed due to the presence of medium value habitat and a significant outlet drop at the culvert. Just upstream of the confluence of this tributary and the Parsnip River, grayling have been documented (Norris [2018] 2024; MoE 2024).

A summary of habitat modelling outputs for the crossing are presented in Table [5.2](#).

Table 5.1: Summary of derived upstream watershed statistics for PSCIS crossing 199663.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
199663	1	713	707	849	784	774	SW

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.2: Summary of fish habitat modelling for PSCIS crossing 199663.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	1.6	1.6	100
BT Spawning (km)	0.7	0.7	100
BT Network (km)	9.9	9.3	94
BT Stream (km)	8.0	7.4	92
BT Lake Reservoir (ha)	0.5	0.5	100
BT Wetland (ha)	12.8	12.8	100
BT Slopeclass03 (km)	3.6	3.0	83
BT Slopeclass05 (km)	3.6	3.6	100
BT Slopeclass08 (km)	0.9	0.9	100
BT Slopeclass15 (km)	0.2	0.2	100

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 199663

At the time of the 2025 assessment, PSCIS crossing 199663 on Chco 12000 was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table [5.3](#)). The culvert had a 0.6m outlet drop and a 0.4m deep outlet pool.

The water temperature was 7°C, pH was 8.2 and conductivity was 310 uS/cm.

Table 5.3: Summary of fish passage assessment for PSCIS crossing 199663.

Location and Stream Data		Crossing Characteristics	
Date	2025-09-21	Crossing Sub Type	Round Culvert
PSCIS ID	199663	Diameter (m)	1.2
External ID	–	Length (m)	13
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	515416	Resemble Channel	–
Northing	6094963	Backwatered	No
Stream	Tributary To Parsnip River	Percent Backwatered	–
Road	Chco 12000	Fill Depth (m)	0.5
Road Tenure	BCTS	Outlet Drop (m)	0.6
Channel Width (m)	0	Outlet Pool Depth (m)	0.4
Stream Slope (%)	1	Inlet Drop	No
Beaver Activity	No	Slope (%)	3
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	30	Barrier Result	Barrier
Fix type	Replace Structure with Streambed Simulation CBS	Fix Span / Diameter	3
Comments: Only photos and downstream width measurements were collected in 2025 due to limited time; all other information was transcribed from the 2024 survey. Habitat confirmations and eDNA samples were collected both upstream and downstream. A large wetland was located approximately 70–80m downstream of the culvert outlet. Abundant gravels were present within the first 20–30m below the outlet pool, where flow was strong, and a substantial outlet drop was observed.			
Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.			

Stream Characteristics at Crossing 19...



Stream Characteristics Downstream of Crossing 199663

The stream was surveyed downstream from crossing 199663 for 80m (Figure 5.2). The habitat was rated as medium value for salmonid spawning and rearing. The stream had good flow and abundant gravels for 30–40m downstream of the culvert. A wetland was located ~80m downstream of the FSR. The average channel width was 2.6m and the average wetted width was 2.8m. The dominant substrate was cobbles with gravels sub-dominant. Total cover amount was rated as abundant with deep pools dominant. Cover was also present as small woody debris, large woody debris, undercut banks, and overhanging vegetation.

Stream Characteristics Upstream of Crossing 199663

The stream was surveyed upstream from crossing 199663 for 600m (Figure 5.2). The habitat was rated as medium value. The dominant substrate was fines with gravels sub-dominant. The average channel width was 3m, the average wetted width was 2.1m, and the average gradient was 2.8%. Total cover amount was rated as abundant with overhanging vegetation dominant. Cover was also present as small woody debris, large woody debris, and undercut banks. The stream was low gradient with significant riparian tree blowdown throughout. Wide floodplain within a well defined canyon (approximately 15m deep with slopes of ~45°). Healthy riparian consisting of herb and shrub cover within sparse spruce forest that appears to have been impacted by some sort of beetle or other affliction. Most of the spruce trees were dead with many already fallen down.

Environmental DNA Sampling

eDNA samples were collected both upstream and downstream of crossing 199663 on Chco 12000, with sampling techniques summarized in Table 5.4.

Table 5.4: Summary of eDNA samples collected at site 199663.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
199663_ds_ed1	Tributary To Parsnip River	riffle	instream	mce	1	2.0	Sample collected 15m downstream of culvert right bank. Site was chosen close to the culvert because of the wetland that was just downstream of this area - and wanted some flow.
199663_us_ed1	Tributary To Parsnip River	riffle	instream	mce	1	2.9	Sample collected 25m upstream of culvert inlet on the right bank

Environmental DNA Results

eDNA samples were collected upstream and downstream of crossing 199663 and submitted to UNBC for ddPCR analysis against Bull Trout, Arctic Grayling, Rainbow Trout, Kokanee, and Burbot assays (Burbot was tested only at this crossing among the Peace 2025 sites). The SOCK ddPCR assay does not distinguish between anadromous Sockeye Salmon and landlocked Kokanee; in Peace tributaries above the Peace Canyon Dam, any SOCK detection would be Kokanee. Per-position results are summarized in Table 5.5. Detections are based on the standard ddPCR call threshold of ≥ 4 positive droplets; format is CODE(max_droplets), with * flagging samples UNBC reran.

Table 5.5: eDNA detection results at crossing 199663.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
199663_ds_ed1	M10139	BURB(195), RAIN(112)	—	BULT(0), GRAY(0)
199663_us_ed1	M10140	RAIN(65)	—	BULT(0), BURB(0), GRAY(0)

Site-level detail across all sampled crossings is in [Appendix - Environmental DNA Results \(page 79\)](#). The [interactive Peace eDNA map](#) shows toggleable per-species layers.

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the Chco 12000 crossing (199663) with a streambed simulation (3m span) is recommended. At the time of reporting in 2026, the cost of the work is estimated at \$NA.

Conclusion

The habitat upstream of PSCIS crossing 199663 on Chco 12000 was documented as medium value for salmonid spawning and rearing, and the crossing is rated as a moderate priority for replacement due to the large outlet drop.

Table 5.6: Summary of habitat details for PSCIS crossing 199663.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
199663	Downstream	80	2.6	2.8	–	–	abundant	–
199663	Upstream	600	3.0	2.1	0.2	2.8	abundant	medium

Conclusion



Figure 5.2: Left: Typical habitat downstream of PSCIS crossing 199663. Right: Typical habitat upstream of PSCIS crossing 199663.

This was the first assessment of PSCIS crossing 203597, in part because the site is situated on a road identified by BC Timber Sales as a potential deactivation candidate. The stream exhibited high-quality habitat, so a habitat confirmation was completed. Bull trout are documented in the Nation River near the confluence with this stream (Norris [2018] 2024; MoE 2024).

A summary of habitat modelling outputs for the crossing are presented in Table [5.8](#).

Table 5.7: Summary of derived upstream watershed statistics for PSCIS crossing 203597.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
203597	8.2	900	907	1065	980	971	S

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.8: Summary of fish habitat modelling for PSCIS crossing 203597.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	5.6	5.6	100
BT Spawning (km)	0.7	0.7	100
BT Network (km)	16.8	9.5	57
BT Stream (km)	12.2	5.2	43
BT Lake Reservoir (ha)	1.0	1.0	100
BT Wetland (ha)	47.1	46.2	98
BT Slopeclass03 (km)	7.9	2.5	32
BT Slopeclass05 (km)	4.0	2.5	62
BT Slopeclass08 (km)	0.5	0.5	100
BT Slopeclass15 (km)	0.0	0.0	–

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 203597

At the time of the 2025 assessment, PSCIS crossing 203597 on Thut 15000 was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table [5.9](#)).

Stream Characteristics at Crossing 20...

The water temperature was 6.5°C, pH was 7.5 and conductivity was 188 uS/cm.

Table 5.9: Summary of fish passage assessment for PSCIS crossing 203597.

Location and Stream Data		Crossing Characteristics	
Date	2025-09-22	Crossing Sub Type	Round Culvert
PSCIS ID	203597	Diameter (m)	0.8
External ID	15201563	Length (m)	19
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	425959	Resemble Channel	–
Northing	6122703	Backwatered	No
Stream	Tributary To Nation River	Percent Backwatered	–
Road	Thut 15000	Fill Depth (m)	1.5
Road Tenure	Sekani R07214 8	Outlet Drop (m)	0.25
Channel Width (m)	2.5	Outlet Pool Depth (m)	0.5
Stream Slope (%)	2.5	Inlet Drop	Yes
Beaver Activity	No	Slope (%)	1
Habitat Value	High	Valley Fill	Deep Fill
Final score	29	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	15
Comments: High-quality stream with abundant gravels and cover, directly connected to the Nation River. Bull trout noted near the confluence. eDNA samples collected upstream and downstream of the crossing. This crossing is located on a road identified by BCTS as a candidate for potential deactivation.			
Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.			

Stream Characteristics at Crossing 20...



Stream Characteristics Downstream of Crossing 203597

The stream was surveyed downstream from crossing 203597 for 400m (Figure 5.4). The stream had good flow, likely elevated due to significant precipitation the previous night. Abundant gravels were present throughout, and cover was provided by multiple sources including woody debris and undercut banks. No barriers were observed. Total cover amount was rated as abundant with undercut banks dominant. Cover was also present as small woody debris, large woody debris, boulders, deep pools, and overhanging vegetation. The average channel width was 2.8m, the average wetted width was 2.6m, and the average gradient was 3%. The dominant substrate was cobbles with gravels sub-dominant.

Stream Characteristics Upstream of Crossing 203597

The stream was surveyed upstream from crossing 203597 for 350m up to the beginning of the wetlands (Figure 5.4). The habitat was rated as NA value. The dominant substrate was fines with cobbles sub-dominant. Total cover amount was rated as moderate with small woody debris dominant. Cover was also present as large woody debris, undercut banks, and overhanging vegetation. The average channel width was 2.5m, the average wetted width was 2.4m, and the average gradient was 3%. The stream had good flow, although recent rainfall may have influenced discharge. Pockets of gravels were present within the area surveyed. Wetlands adjacent to the channel appeared shallow, suggesting they may freeze to the bottom during winter.

Environmental DNA Sampling

eDNA samples were collected both upstream and downstream of crossing 203597 on Thut 15000, with sampling techniques summarized in Table 5.10.

Table 5.10: Summary of eDNA samples collected at site 203597.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
203597_ds_ed1b	Tributary To Nation River	riffle	grab	mce	1	1.7	Sample collected 350 - 400m downstream of culvert. Right bank. Accessed via historic road which is now wildlife trail.
203597_us_ed1	Tributary To Nation River	riffle	inatream	mce	1	2.0	Sample collected 5m upstream of the culvert on the left bank

Environmental DNA Results

eDNA samples were collected upstream and downstream of crossing 203597 and submitted to UNBC for ddPCR analysis against Bull Trout, Arctic Grayling, Rainbow Trout, and Kokanee assays (the SOCK ddPCR assay does not distinguish between anadromous Sockeye Salmon and landlocked Kokanee; in Peace tributaries above the Peace Canyon Dam, any SOCK detection would be Kokanee). Per-position results from real environmental samples are summarized in Table 5.11. Detections are based on the standard ddPCR call threshold of ≥ 4 positive droplets; format is CODE(max_droplets), with * flagging samples UNBC reran.

Table 5.11: eDNA detection results at crossing 203597 (real environmental samples).

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
203597_ds_ed1b	M41366	RAIN(173)*	BULT(3)*	GRAY(0), SOCK(0)
203597_us_ed1	M41367	RAIN(213)*	—	BULT(0), GRAY(0), SOCK(0)*

Field blank at this crossing. Sample 203597_ds_ed1a (UNBC ID M41365) was a field blank — distilled water filtered at the field site to test for protocol contamination, not site eDNA. UNBC's analysis returned a low-level Rainbow Trout signal at this blank (4 positive droplets, ≈ 0.27 copies/ μL). Per ddPCR convention this represents protocol-side contamination during collection or processing, not species presence at this location. The blank is presented in the Controls layer of the interactive map (off by default) and in the field-blanks table within [Appendix - Environmental DNA Results \(page 79\)](#).

Site-level detail across all sampled crossings is in [Appendix - Environmental DNA Results \(page 79\)](#). The [interactive Peace eDNA map](#) shows toggleable per-species layers.

Structure Remediation and Cost Estimate

Ideally, this crossing is deactivated by BC Timber Sales; however, if restoration or maintenance activities proceed, replacement of the Thut 15000 crossing (203597) with a bridge (15m span) is recommended. At the time of reporting in 2026, the cost of the work is estimated at \$450,000.

Conclusion

Tributary To Nation River - 203597 - A...

by BC Timber Sales.

Conclusion

Table 5.12: Summary of habitat details for PSCIS crossing 203597.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
203597	Downstream	400	2.8	2.6	0.3	3	abundant	high
203597	Upstream	350	2.5	2.4	0.3	3	moderate	–



Figure 5.4: Left: Typical habitat downstream of PSCIS crossing 203597. Right: Wetland habitat located 350m upstream of PSCIS crossing 203597.

Tributary To Williston Reservoir - 203605 - Appendix

Site Location

PSCIS crossing 203605 is located on Tributary To Williston Reservoir, on the western side of the reservoir, approximately 10km southeast of MacKenzie, BC, in the Parsnip Arm watershed group (Figure 5.5). The crossing is located 1.3m upstream of the stream's confluence with the Williston Reservoir on Finlay FSR road which is the responsibility of the Ministry of Forests.

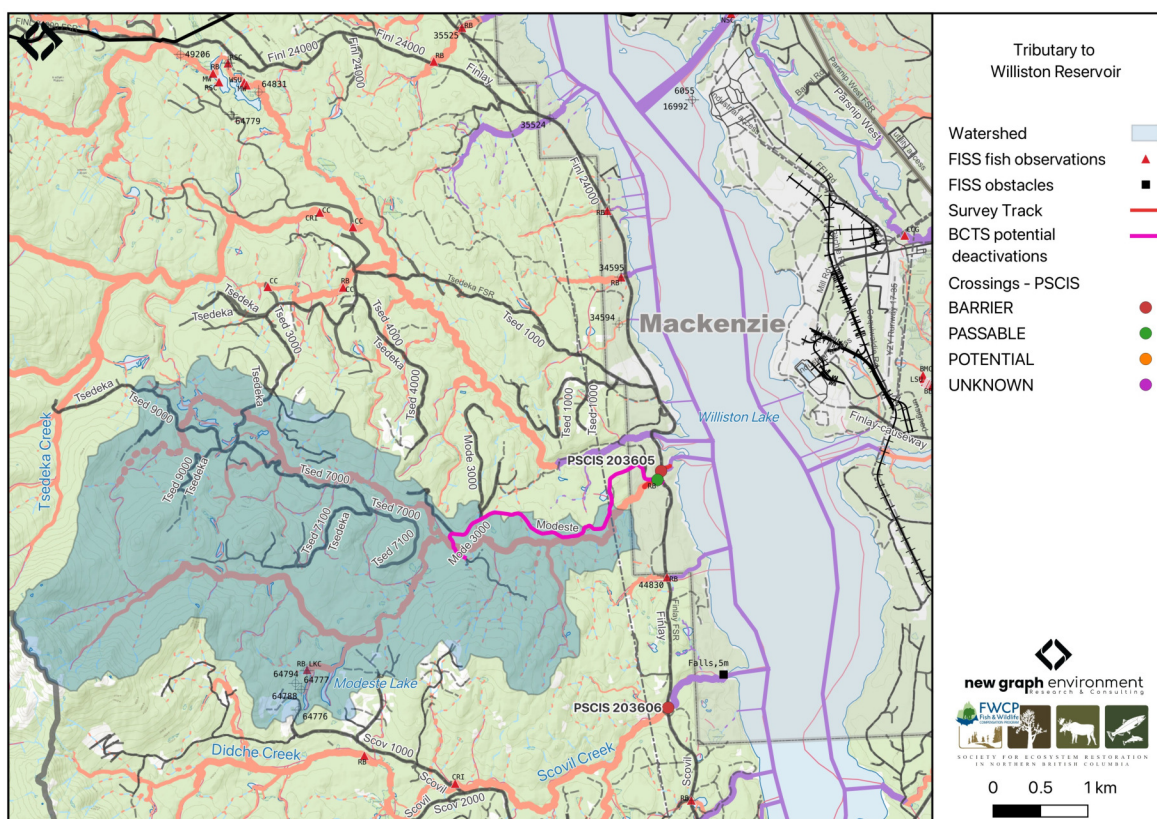


Figure 5.5: Map of Tributary To Williston Reservoir

Background

At PSCIS crossing 203605, Tributary To Williston Reservoir is a fourth order stream and drains a watershed of approximately 55.6km². The watershed ranges in elevation from a maximum of 1644m to 681m near the crossing (Table 5.13).

This was the first assessment of PSCIS crossing 203605, selected because it is located very close to Williston Reservoir, which supports documented bull trout. Approximately 35km of bull trout rearing habitat is modelled upstream with no additional modelled barriers, and lake chub and rainbow trout have previously been recorded in the system (Norris [2018] 2024; MoE 2024). The stream exhibited high-quality habitat, therefore a habitat confirmation was completed.

Approximately 215m upstream of Finlay FSR, the stream crosses Modeste FSR at a bridge, which was also assessed for the first time in 2025 (PSCIS crossing 203611). Modeste FSR provides access to the lower half of the watershed, and is identified by BC Timber Sales as a potential deactivation candidate.

A summary of habitat modelling outputs for the crossing are presented in Table [5.14](#).

Table 5.13: Summary of derived upstream watershed statistics for PSCIS crossing 203605.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
203605	55.6	681	671	1644	980	937	SE

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.14: Summary of fish habitat modelling for PSCIS crossing 203605.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	35.9	30.8	86
BT Spawning (km)	22.0	19.5	89
BT Network (km)	100.5	76.4	76
BT Stream (km)	86.7	64.1	74
BT Lake Reservoir (ha)	51.1	42.4	83
BT Wetland (ha)	51.9	50.3	97
BT Slopeclass03 (km)	16.3	14.4	88
BT Slopeclass05 (km)	21.8	10.4	48
BT Slopeclass08 (km)	17.6	12.2	69
BT Slopeclass15 (km)	28.2	24.4	87

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 203605

At the time of the 2025 assessment, PSCIS crossing 203605 on Finlay FSR was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table [5.15](#)). The culvert had a 0.95m outlet drop and a 0.5m deep outlet pool.

The water temperature was 6.8°C, pH was 7.6 and conductivity was 251 uS/cm.

Table 5.15: Summary of fish passage assessment for PSCIS crossing 203605.

Location and Stream Data		Crossing Characteristics	
Date	2025-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	203605	Diameter (m)	3.9
External ID	16401519	Length (m)	46
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	486504	Resemble Channel	–
Northing	6124990	Backwatered	No
Stream	Tributary To Williston Reservoir	Percent Backwatered	–
Road	Finlay FSR	Fill Depth (m)	2.5
Road Tenure	MoF	Outlet Drop (m)	0.95
Channel Width (m)	6	Outlet Pool Depth (m)	0.5
Stream Slope (%)	2	Inlet Drop	No
Beaver Activity	No	Slope (%)	2
Habitat Value	High	Valley Fill	Deep Fill
Final score	37	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	15
Comments: High value habitat on very large system with pipe that has large outlet drop near 1m. Habitat confirmation and eDNA sampling conducted upstream and downstream of the crossing.			
Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.			

Stream Characteristics at Crossing 20...



Stream Characteristics Downstream of Crossing 203605

The stream was surveyed downstream from crossing 203605 for 250m to the mapped reservoir location, where the channel retained stream characteristics at the time of assessment (Figure 5.6). Abundant gravels and cover were present throughout, with numerous run sections providing suitable rearing habitat for juvenile westslope cutthroat trout and rainbow trout. No barriers to fish passage were observed in the area surveyed and the habitat was rated as high value for salmonid spawning and rearing. The average channel width was 8.5m, the average wetted width was 5.6m, and the average gradient was 1.8%. Total cover amount was rated as moderate with dominant. Cover was also present as large woody debris and overhanging vegetation. The dominant substrate was gravels with fines sub-dominant.

Stream Characteristics Upstream of Crossing 203605

The stream was surveyed upstream from crossing 203605 for 650m (Figure 5.6). The habitat was rated as high value. Total cover amount was rated as moderate with undercut banks dominant. Cover was also present as small woody debris, large woody debris, boulders, deep pools, and overhanging vegetation. The average channel width was 6.7m, the average wetted width was 4.6m, and the average gradient was 2.9%. The dominant substrate was gravels with cobbles sub-dominant. The stream was large with complex habitat and diverse cover including deep pools, boulders, large and small woody debris, and undercut banks. A canyon section ~100m long occurred ~100m upstream of the Modeste FSR bridge, containing multiple cascade steps up to 1m high with deep downstream pools. At the upstream end of the site, the channel deflected against steep gully walls. Abundant gravels were present, suitable for adfluvial and resident rainbow trout as well as bull trout.

Environmental DNA Sampling

eDNA samples were collected both upstream and downstream of crossing 203605 on Finlay FSR, with sampling techniques summarized in Table 5.16.

Table 5.16: Summary of eDNA samples collected at site 203605.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
203605_ds_ed1	Tributary To Williston Reservoir	riffle	grab	mce	1	–	Sample collected 15m downstream of the culvert outlet on the right bank
203605_us_ed1	Tributary To Williston Reservoir	riffle	instream	mce	1	2	Approximately 45 m upstream of culvert inlet on the left bank

Environmental DNA Results

eDNA samples were collected upstream and downstream of crossing 203605 and submitted to UNBC for ddPCR analysis against Bull Trout, Arctic Grayling, Rainbow Trout, and Kokanee assays (the SOCK ddPCR assay does not distinguish between anadromous Sockeye Salmon and landlocked Kokanee; in Peace tributaries above the Peace Canyon Dam, any SOCK detection would be Kokanee). Per-position results are summarized in Table 5.17. Detections are based on the standard ddPCR call threshold of ≥ 4 positive droplets; format is CODE(max_droplets), with * flagging samples UNBC reran. Sub-threshold signals (1-3 droplets) are tracked separately as they are not interpretable as species presence per ddPCR convention.

Table 5.17: eDNA detection results at crossing 203605.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
203605_ds_ed1	M41368	RAIN(455)	BULT(3)*	GRAY(0), SOCK(0)
203605_us_ed1	M41369	RAIN(429)	BULT(1)	GRAY(0), SOCK(0)

Site-level detail across all sampled crossings is in [Appendix - Environmental DNA Results \(page 79\)](#). The [interactive Peace eDNA map](#) shows toggleable per-species layers.

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the Finlay FSR crossing (203605) with a bridge (15m span) is recommended. At the time of reporting in 2026, the cost of the work is estimated at \$450,000.

Conclusion

The habitat upstream of PSCIS crossing 203605 on Finlay FSR was documented as high value for salmonid spawning and rearing, and the crossing is rated as a high priority for replacement due to the large outlet drop. There is a significant amount of bull trout rearing habitat modelled upstream, and since the road that provides access to lower half of the may be deactivated, this crossing could be a good candidate for replacement.

Table 5.18: Summary of habitat details for PSCIS crossing 203605.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
203605	Downstream	250	8.5	5.6	0.2	1.8	moderate	high
203605	Upstream	650	6.7	4.6	3.7	2.9	moderate	high

Conclusion



Figure 5.6: Left: Typical habitat downstream of PSCIS crossing 203605. Right: Typical habitat upstream of PSCIS crossing 203605.

Appendix - Environmental DNA Results

This appendix contains site-level detail for the 2025 environmental DNA (eDNA) sampling program in the Peace project area. Headline per-species detection counts are presented in the main Results chapter; the tables below show which sites detected which species. Field blanks are tracked separately as quality-assurance samples — they test for protocol contamination, not site eDNA. Office blanks (samples filtered at the field accommodation rather than at site) are excluded from these tables because their UTM coordinates are inherited from related sites and do not represent real sampling locations.

The interactive map of these results is linked from the main Results chapter and at the end of this appendix.

Per-site detection results

Format: CODE(max_droplets), with * flagging samples that UNBC reran. Detected = ≥ 4 positive droplets per the standard ddPCR call threshold; Sub-threshold = 1-3 droplets, not interpretable as species presence per ddPCR convention; Not detected = 0 droplets across all runs.

Table 5.19: Site-level eDNA detection results across real environmental samples in the Peace project area. UNBC ID is the lab sample identifier; Pos. control = TRUE indicates the sample was collected at a site where the target species was also confirmed by electrofishing.

Site ID	Stream	UNBC ID	Pos. control	Detected	Sub-threshold	Not detected
125000_ds_ed1	Tributary To Parsnip River	M41347		RAIN(980)	—	BULT(0), GRAY(0), SOCK(0)
125000_us_ed1	Tributary To Parsnip River	M41348	—	RAIN(143)	—	BULT(0), GRAY(0), SOCK(0)
125179_ds_ed1a	Tributary To Missinka River	M41349	—	RAIN(309)	BULT(1)*	GRAY(0), SOCK(0)
125179_us_ed1a	Tributary To Missinka River	M41351	yes	RAIN(383)*	—	BULT(0), GRAY(0), SOCK(0)
125180_ds_ed1	Tributary To Missinka River	M41353		BULT(4), RAIN(330)	—	GRAY(0), SOCK(0)
125180_us_ed1	Tributary To Missinka River	M41354		RAIN(293)	BULT(3)	GRAY(0), SOCK(0)
125194_ds_ed1a	Tributary To Missinka River	M21242		—	BULT(1), RAIN(3)*	—
125231_ds_ed1c	Tributary To Table River	M41356	—	RAIN(205)	—	BULT(0), GRAY(0), SOCK(0)
125231_us_ed1	Tributary To Table River	M41357	—	RAIN(5)	—	BULT(0), GRAY(0), SOCK(0)

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Site ID	Stream	UNBC ID	Pos. control	Detected	Sub-threshold	Not detected
125261_ds_ed1	Fern Creek	M41358	—	RAIN(258)	—	BULT(0), GRAY(0), SOCK(0)
125261_us_ed1a	Fern Creek	M41359	—	RAIN(533)	—	BULT(0), GRAY(0), SOCK(0)
125261_us_ed1b	Fern Creek	M41360	—	RAIN(567)	—	BULT(0), GRAY(0), SOCK(0)
198692_ds_ed1b	Tributary To Kerry Lake	M41362	—	RAIN(119)	BULT(1)	GRAY(0), SOCK(0)
198692_ds_ed2	Tributary To Kerry Lake	M41363	—	RAIN(630)	—	BULT(0), GRAY(0), SOCK(0)
198692_us_ed1	Tributary To Kerry Lake	M41364	—	RAIN(235)	—	BULT(0), GRAY(0), SOCK(0)
199663_ds_ed1	Tributary To Parsnip River	M10139	—	BURB(195), RAIN(112)	—	BULT(0), GRAY(0)
199663_us_ed1	Tributary To Parsnip River	M10140	—	RAIN(65)	—	BULT(0), BURB(0), GRAY(0)
203597_ds_ed1b	Tributary To Nation River	M41366		RAIN(173)*	BULT(3)*	GRAY(0), SOCK(0)
203597_us_ed1	Tributary To Nation River	M41367		RAIN(213)*	—	BULT(0), GRAY(0), SOCK(0)*
203605_ds_ed1	Tributary To Williston Reservoir	M41368	—	RAIN(455)	BULT(3)*	GRAY(0), SOCK(0)
203605_us_ed1	Tributary To Williston Reservoir	M41369	—	RAIN(429)	BULT(1)	GRAY(0), SOCK(0)
24718557_ed1	Missinka River	M41370	—	RAIN(4)	BULT(2), GRAY(1)	SOCK(0)
57690_ds_ed1	Tributary To Wichcika Creek	M41371	—	RAIN(10)	—	BULT(0), GRAY(0), SOCK(0)
57690_us_ed1	Tributary To Wichcika Creek	M41372	—	—	—	BULT(0), GRAY(0), RAIN(0), SOCK(0)
57695_ds_ed1	Tributary To Wichcika Creek	M41373	—	—	—	BULT(0), GRAY(0), RAIN(0), SOCK(0)
57695_us_ed1	Tributary To Wichcika Creek	M21244	—	—	RAIN(1)	BULT(0)
57696_ds_ed1	Tributary To Wichcika Creek	M21245		—	—	BULT(0), RAIN(0)
57696_us_ed1	Tributary To Wichcika Creek	M21246		—	—	BULT(0), RAIN(0)

Field blanks

Field blanks filter distilled water at the field site to test for protocol contamination during collection and processing. **Detections in blanks indicate potential cross-contamination, not species presence at the location** — these rows are presented for QA transparency only, not as eDNA detections.

Retests

Table 5.20: Field blanks collected during the 2025 eDNA program in the Peace project area. Detections in blanks reflect protocol-side contamination, not site eDNA.

Site ID	Stream	UNBC ID	Detected	Sub-threshold	Clean (0 droplets)
125231_ds_ed1a	Tributary To Table River	M41355	—	—	BULT(0), GRAY(0), RAIN(0), SOCK(0)
198692_ds_ed1a	Tributary To Kerry Lake	M41361	RAIN(5)*	—	BULT(0), GRAY(0), SOCK(0)*
203597_ds_ed1a	Tributary To Nation River	M41365	RAIN(4)*	—	BULT(0), GRAY(0), SOCK(0)*

Retests

UNBC reran a subset of samples to confirm borderline results. The table below lists every (site x species) combination that was rerun, including reruns from both real environmental samples and field blanks.

Table 5.21: UNBC retests for the 2025 Peace eDNA samples. Runs is the total number of times the sample x species combination was assayed; Max droplets is the highest positive-droplet count observed across all runs.

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
125179_ds_ed1a	Tributary To Missinka River	Bull Trout	BULT	environmental	4	1	Sub-threshold (1-3 droplets, NOT a detection)
125179_us_ed1a	Tributary To Missinka River	Rainbow Trout	RAIN	environmental	4	383	Detected (≥ 4 droplets)
125194_ds_ed1a	Tributary To Missinka River	Rainbow Trout	RAIN	environmental	4	3	Sub-threshold (1-3 droplets, NOT a detection)
198692_ds_ed1a	Tributary To Kerry Lake	Bull Trout	BULT	field blank	4	0	No detection (0 droplets)
198692_ds_ed1a	Tributary To Kerry Lake	Arctic Grayling	GRAY	field blank	4	0	No detection (0 droplets)
198692_ds_ed1a	Tributary To Kerry Lake	Rainbow Trout	RAIN	field blank	6	5	Detected (≥ 4 droplets)
198692_ds_ed1a	Tributary To Kerry Lake	Sockeye Salmon	SOCK	field blank	6	0	No detection (0 droplets)
203597_ds_ed1a	Tributary To Nation River	Bull Trout	BULT	field blank	4	0	No detection (0 droplets)
203597_ds_ed1a	Tributary To Nation River	Arctic Grayling	GRAY	field blank	4	0	No detection (0 droplets)
203597_ds_ed1a	Tributary To Nation River	Rainbow Trout	RAIN	field blank	4	4	Detected (≥ 4 droplets)

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Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
203597_ds_ed1a	Tributary To Nation River	Sockeye Salmon	SOCK	field blank	4	0	No detection (0 droplets)
203597_ds_ed1b	Tributary To Nation River	Bull Trout	BULT	environmental	4	3	Sub-threshold (1-3 droplets, NOT a detection)
203597_ds_ed1b	Tributary To Nation River	Arctic Grayling	GRAY	environmental	4	0	No detection (0 droplets)
203597_ds_ed1b	Tributary To Nation River	Rainbow Trout	RAIN	environmental	4	173	Detected (≥ 4 droplets)
203597_ds_ed1b	Tributary To Nation River	Sockeye Salmon	SOCK	environmental	4	0	No detection (0 droplets)
203597_us_ed1	Tributary To Nation River	Bull Trout	BULT	environmental	4	0	No detection (0 droplets)
203597_us_ed1	Tributary To Nation River	Arctic Grayling	GRAY	environmental	4	0	No detection (0 droplets)
203597_us_ed1	Tributary To Nation River	Rainbow Trout	RAIN	environmental	4	213	Detected (≥ 4 droplets)
203597_us_ed1	Tributary To Nation River	Sockeye Salmon	SOCK	environmental	4	0	No detection (0 droplets)
203605_ds_ed1	Tributary To Williston Reservoir	Bull Trout	BULT	environmental	4	3	Sub-threshold (1-3 droplets, NOT a detection)
24718557_ed1	Missinka River	Bull Trout	BULT	environmental	4	2	Sub-threshold (1-3 droplets, NOT a detection)
24718557_ed1	Missinka River	Arctic Grayling	GRAY	environmental	4	1	Sub-threshold (1-3 droplets, NOT a detection)

Interactive map

The interactive Peace eDNA map shows site-level detection results with toggleable per-species layers, an opt-in sub-threshold-signals layer (off by default), and an opt-in Controls layer for field blanks. See the [interactive Peace eDNA map](#).

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Note that this reference was not actually cited in the body of the report but rather in the `index.Rmd` file `yam1 headerj` under the `no_cite` key.

Session Info

Information about the computing environment is important for reproducibility. A summary of the computing environment is saved to `session_info.csv`, which can be viewed and downloaded from [here](#).

– Session info

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## setting value
## version R version 4.5.2 (2025-10-31)
## os      macOS Tahoe 26.2
## system  aarch64, darwin20
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## language (EN)
## collate en_US.UTF-8
## ctype   en_US.UTF-8
## tz      America/Vancouver
## date    2026-05-04
## pandoc  3.0.1 @ /opt/homebrew/bin/ (via rmarkdown)
## quarto  1.4.553 @ /usr/local/bin/quarto
##
```

– Packages

## package	* version	date (UTC)	lib	source
## arrow	22.0.0.1	2025-12-23	[2]	CRAN (R 4.5.2)
## askpass	1.2.1	2024-10-04	[1]	CRAN (R 4.5.0)
## assertthat	0.2.1	2019-03-21	[2]	CRAN (R 4.5.0)
## base64enc	0.1-6	2026-02-02	[1]	CRAN (R 4.5.2)
## bit	4.6.0	2025-03-06	[1]	CRAN (R 4.5.0)
## bit64	4.8.0	2026-04-21	[1]	CRAN (R 4.5.2)
## blob	1.3.0	2026-01-14	[1]	CRAN (R 4.5.2)
## bookdown	* 0.46	2025-12-05	[2]	CRAN (R 4.5.2)
## bslib	0.10.0	2026-01-26	[1]	CRAN (R 4.5.2)
## cachem	1.1.0	2024-05-16	[1]	CRAN (R 4.5.0)
## chk	0.10.0	2025-01-24	[1]	CRAN (R 4.5.0)
## class	7.3-23	2025-01-01	[2]	CRAN (R 4.5.2)
## classInt	0.4-11	2025-01-08	[1]	CRAN (R 4.5.0)
## cli	3.6.6	2026-04-09	[1]	CRAN (R 4.5.2)
## codetools	0.2-20	2024-03-31	[2]	CRAN (R 4.5.2)
## crayon	1.5.3	2024-06-20	[1]	CRAN (R 4.5.0)
## crosstalk	1.2.2	2025-08-26	[1]	CRAN (R 4.5.0)
## curl	7.1.0	2026-04-22	[1]	CRAN (R 4.5.2)
## DBI	1.3.0	2026-02-25	[1]	CRAN (R 4.5.2)
## dbplyr	2.5.2	2026-02-13	[1]	CRAN (R 4.5.2)
## devtools	2.4.6	2025-10-03	[2]	CRAN (R 4.5.0)
## digest	0.6.39	2025-11-19	[1]	CRAN (R 4.5.2)

Session Info

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## e1071                1.7-17      2025-12-18 [1] CRAN (R 4.5.2)
## elevatr              0.99.1      2025-09-10 [2] CRAN (R 4.5.0)
## ellipsis             0.3.2       2021-04-29 [2] CRAN (R 4.5.0)
## english              1.2-6       2021-08-21 [2] CRAN (R 4.5.0)
## evaluate             1.0.5       2025-08-27 [1] CRAN (R 4.5.0)
## farver               2.1.2       2024-05-13 [1] CRAN (R 4.5.0)
## fasstr               0.5.3       2024-09-27 [2] CRAN (R 4.5.0)
## fastmap              1.2.0       2024-05-15 [1] CRAN (R 4.5.0)
## fishbc               * 0.2.1.9000 2026-01-15 [2] Github (lucy-
schick/fishbc@0607ffd)
## forcats              * 1.0.1      2025-09-25 [2] CRAN (R 4.5.0)
## foreach              1.5.2       2022-02-02 [2] CRAN (R 4.5.0)
## fpr                  * 1.2.0      2026-02-03 [2] Github
(NewGraphEnvironment/fpr@2817533)
## fs                   2.1.0       2026-04-18 [1] CRAN (R 4.5.2)
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## gh                   1.5.0       2025-05-26 [2] CRAN (R 4.5.0)
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## htmlwidgets          1.6.4       2023-12-06 [1] CRAN (R 4.5.0)
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## janitor              2.2.1       2024-12-22 [1] CRAN (R 4.5.0)
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## KernSmooth           2.23-26     2025-01-01 [2] CRAN (R 4.5.2)
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## lattice              0.22-9      2026-02-09 [2] CRAN (R 4.5.2)
## leafem               * 0.2.5       2025-08-28 [2] CRAN (R 4.5.0)
## leaflet              * 2.2.3       2025-09-04 [1] CRAN (R 4.5.0)
## lifecycle            1.0.5       2026-01-08 [1] CRAN (R 4.5.2)
## lubridate            * 1.9.5       2026-02-04 [1] CRAN (R 4.5.2)
## magick               2.9.0       2025-09-08 [2] CRAN (R 4.5.0)
## magrittr             2.0.5       2026-04-04 [1] CRAN (R 4.5.2)
## memoise              2.0.1       2021-11-26 [1] CRAN (R 4.5.0)
## ngr                  * 0.0.1       2026-01-22 [2] Github
(newgraphenvironment/ngr@22c9714)
## otel                 0.2.0       2025-08-29 [1] CRAN (R 4.5.0)
```

```

## pagedown          * 0.23      2025-08-20 [2] CRAN (R 4.5.0)
## pak                0.9.5      2026-04-27 [1] CRAN (R 4.5.2)
## pdftools          * 3.7.0      2026-01-30 [2] CRAN (R 4.5.2)
## pillar            1.11.1      2025-09-17 [1] CRAN (R 4.5.0)
## pkgbuild           1.4.8      2025-05-26 [2] CRAN (R 4.5.0)
## pkgconfig          2.0.3      2019-09-22 [1] CRAN (R 4.5.0)
## pkgload            1.5.0      2026-02-03 [2] CRAN (R 4.5.2)
## plyr               1.8.9      2023-10-02 [1] CRAN (R 4.5.0)
## png                0.1-9      2026-03-15 [1] CRAN (R 4.5.2)
## prettyunits        1.2.0      2023-09-24 [1] CRAN (R 4.5.0)
## processx           3.9.0      2026-04-22 [1] CRAN (R 4.5.2)
## progress           1.2.3      2023-12-06 [2] CRAN (R 4.5.0)
## progressr          0.18.0     2025-11-06 [2] CRAN (R 4.5.0)
## proxy              0.4-29      2025-12-29 [1] CRAN (R 4.5.2)
## purrr              * 1.2.2      2026-04-10 [1] CRAN (R 4.5.2)
## qpdf               1.4.1      2025-07-02 [2] CRAN (R 4.5.0)
## R6                 2.6.1      2025-02-15 [1] CRAN (R 4.5.0)
## rappdirs           0.3.4      2026-01-17 [1] CRAN (R 4.5.2)
## raster             3.6-32      2025-03-28 [1] CRAN (R 4.5.0)
## rayshader          0.37.3      2024-02-21 [2] CRAN (R 4.5.0)
## rbbt               * 0.0.0.9000 2026-01-15 [2] Github
(paleolimbot/rbbt@212680c)
## RColorBrewer       1.1-3      2022-04-03 [1] CRAN (R 4.5.0)
## Rcpp               1.1.1-1.1 2026-04-24 [1] CRAN (R 4.5.2)
## RcppRoll           0.3.1      2024-07-07 [2] CRAN (R 4.5.0)
## readr              * 2.2.0      2026-02-19 [1] CRAN (R 4.5.2)
## readwritesqlite * 0.2.0.9021 2026-01-15 [2] Github
(poissonconsulting/readwritesqlite@ccdb297)
## remotes            2.5.0      2024-03-17 [2] CRAN (R 4.5.0)
## rgl                1.3.31     2025-11-14 [2] CRAN (R 4.5.2)
## rlang              1.2.0      2026-04-06 [1] CRAN (R 4.5.2)
## rmarkdown          * 2.31      2026-03-26 [1] CRAN (R 4.5.2)
## RPostgres          * 1.4.10     2026-02-16 [1] CRAN (R 4.5.2)
## rprojroot          2.1.1      2025-08-26 [2] CRAN (R 4.5.0)
## RSQLite            2.4.6      2026-02-06 [2] CRAN (R 4.5.2)
## rstudioapi         0.18.0     2026-01-16 [2] CRAN (R 4.5.2)
## rvest              1.0.5      2025-08-29 [2] CRAN (R 4.5.0)
## s2                 1.1.9      2025-05-23 [1] CRAN (R 4.5.0)
## S7                 0.2.2      2026-04-22 [1] CRAN (R 4.5.2)
## sass               0.4.10     2025-04-11 [1] CRAN (R 4.5.0)
## scales             1.4.0      2025-04-24 [1] CRAN (R 4.5.0)
## sessioninfo        1.2.3      2025-02-05 [2] CRAN (R 4.5.0)
## sf                 * 1.1-0      2026-02-24 [1] CRAN (R 4.5.2)
## snakecase          0.11.1     2023-08-27 [1] CRAN (R 4.5.0)
## sp                 2.2-1      2026-02-13 [1] CRAN (R 4.5.2)
## staticimports      * 0.0.0.9001 2026-01-15 [2] Github
(newgraphenvironment/staticimports@19cd73c)

```

Session Info

```
## stringi          1.8.7      2025-03-27 [1] CRAN (R 4.5.0)
## stringr          * 1.6.0      2025-11-04 [1] CRAN (R 4.5.0)
## svglite          2.2.2      2025-10-21 [2] CRAN (R 4.5.0)
## systemfonts      1.3.1      2025-10-01 [2] CRAN (R 4.5.0)
## terra            1.9-11     2026-03-26 [1] CRAN (R 4.5.2)
## textshaping       1.0.4      2025-10-10 [2] CRAN (R 4.5.0)
## tibble           * 3.3.1      2026-01-11 [1] CRAN (R 4.5.2)
## tidyhydat         0.7.2      2025-10-23 [2] CRAN (R 4.5.0)
## tidyr            * 1.3.2      2025-12-19 [1] CRAN (R 4.5.2)
## tidyselect        1.2.1      2024-03-11 [1] CRAN (R 4.5.0)
## tidyverse         * 2.0.0      2023-02-22 [2] CRAN (R 4.5.0)
## tidyxl           1.0.10     2026-01-15 [2] Github
(nacnudus/tidyxl@7e2fbe7)
## timechange        0.4.0      2026-01-29 [1] CRAN (R 4.5.2)
## tzdb              0.5.0      2025-03-15 [1] CRAN (R 4.5.0)
## units             1.0-1      2026-03-11 [1] CRAN (R 4.5.2)
## usethis           3.2.1      2025-09-06 [2] CRAN (R 4.5.0)
## vctrs             0.7.3      2026-04-11 [1] CRAN (R 4.5.2)
## viridisLite        0.4.3      2026-02-04 [1] CRAN (R 4.5.2)
## vroom             1.7.1      2026-03-31 [1] CRAN (R 4.5.2)
## withr             3.0.2      2024-10-28 [1] CRAN (R 4.5.0)
## wk                0.9.5      2025-12-18 [1] CRAN (R 4.5.2)
## xfun              0.57       2026-03-20 [1] CRAN (R 4.5.2)
## xml2              1.5.2      2026-01-17 [1] CRAN (R 4.5.2)
## yaml              2.3.12     2025-12-10 [1] CRAN (R 4.5.2)
##
## [1] /Users/airvine/Library/R/arm64/4.5/library
## [2] /Library/Frameworks/R.framework/Versions/4.5-
arm64/Resources/library
## * — Packages attached to the search path.
##
##
```

Attachment - Maps

All georeferenced field maps are presented at:

- <https://hillcrestgeo.ca/outgoing/fishpassage/projects/parsnip/archive/2022-05-27/>

Maps are also available zipped for bulk download at:

- <https://hillcrestgeo.ca/outgoing/fishpassage/projects/parsnip/archive/2022-05-27/2022-05-27.zip>

Attachment - Phase 1 Data and Photos

Data and photos for all Phase 1 - fish passage assessments are provided online at https://www.newgraphenvironment.com/fish_passage_peace_2024_reporting/appendix---phase-1-fish-passage-assessment-data-and-photos.html - with a pdf version available at https://github.com/NewGraphEnvironment/fish_passage_peace_2024_reporting/raw/main/docs/Appendix_1.pdf

Attachment - Habitat Assessment and Fish Sampling Data

All field data collected is available [here](#).

Habitat assessment data (including fish sampling and PIT tagging information) is available for download here https://github.com/NewGraphEnvironment/fish_passage_peace_2025_reporting/blob/main/data/habitat_confirmations.xls.

Raw fish data is available for download here https://github.com/NewGraphEnvironment/fish_passage_peace_2025_reporting/blob/main/data/fish_data_tags_joined.csv.

Attachment - Bayesian analysis to map stream discharge and temperature causal effects pathways

Details of this analysis and subsequent outputs can be reviewed in the report [Spatial Stream Network Analysis of Nechako Watershed Stream Temperatures 2022b](#) (Hill, Thorley, and Irvine 2024). At the time of reporting, ongoing work regarding the project was tracked <https://github.com/poissonconsulting/fish-passage-22/issues> and <https://github.com/poissonconsulting/fish-passage-22b/issues>.